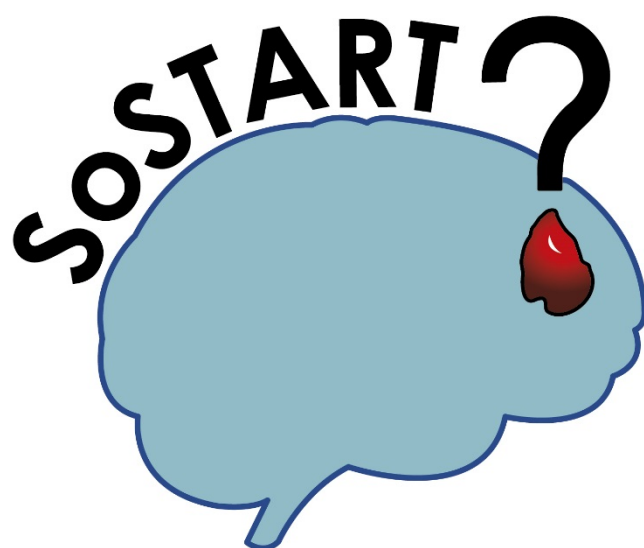




Start or STop Anticoagulants Randomised Trial

ADMINISTRATIVE INFORMATION



THE UNIVERSITY of EDINBURGH



TRIAL PROTOCOL¹

Title	Start or STop Anticoagulants Randomised Trial (SoSTART) after spontaneous intracranial haemorrhage
Co-sponsors	University of Edinburgh & NHS Lothian ACCORD The Queen's Medical Research Institute 47 Little France Crescent Edinburgh EH16 4TJ
Funders	Chest Heart & Stroke Scotland (Res16/A166), Medical Research Council (G1002605), and British Heart Foundation (CS/18/2/33719).
Chief investigator	Professor Rustam Al-Shahi Salman
Sponsor number	AC16141
EudraCT number	2016-004121-16
REC number	17/SS/0082
ClinicalTrials.gov Identifier	NCT03153150
Version number and date	6.0, 23 January 2020

¹ This protocol has been reported according to the SPIRIT 2013 reporting guidelines (Chan AW et al. SPIRIT 2013 statement: defining standard protocol items for clinical trials. *Ann Intern Med* 2013;158(3):200-7)

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Role of the sponsor

The sponsor did not have any role in the design of the trial. They will not have any role in the collection, management, analysis, and interpretation of data. They will not be involved in writing the report of the trial, or the decision to submit the report for publication, and they will not have ultimate authority over any of these activities.

Role of the funders

The funders did not have any role in the design of the trial. They will not have any role in the collection, management, analysis, and interpretation of data. They will not be involved in writing the report of the trial, or the decision to submit the report for publication, and they will not have ultimate authority over any of these activities.

Committee composition

The composition, roles, and responsibilities of the trial coordinating centre, trial steering committee, end point adjudication committee, data management team, and other individuals or groups overseeing the trial are listed in section 25 below on page 65 and 26 below on page 66.

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1 PROTOCOL APPROVAL

1.1 TRIAL NAME

Start or STop Anticoagulants Randomised Trial (SoSTART) after spontaneous intracranial haemorrhage

1.2 EUDRACT NUMBER

2016-004121-16

1.3 SIGNATURES

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Chief Investigator

Signature

Date

Steff Lewis
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Signature

Date

Fiach O'Mahony
**Sponsor(s)
Representative**

Signature

Date

Principal Investigator

Signature

Date

2 LIST OF ABBREVIATIONS

ACCORD	Academic and Clinical Central Office for Research & Development - Joint office for University of Edinburgh and NHS Lothian
AE	Adverse event
AF	Atrial fibrillation/flutter (persistent or paroxysmal)
AR	Adverse reaction
CRF	Case report form
CTIMP	Clinical trial of an investigational medicinal product
DMC	Data monitoring committee
DOAC	Direct oral anticoagulant
eCRF	Electronic case report form
GCP	Good clinical practice
GP	General (family/primary care) practitioner
ICF	Informed consent form
ICH	Intracerebral haemorrhage
IMP	Investigational medicinal product
ISF	Investigator site file
INR	International normalised ratio
IVH	Intraventricular haemorrhage
mRS	modified Rankin scale
OAC	Oral anticoagulation
pCRF	Paper case report form
PIL	Patient information leaflet
PLR	Personal legal representative
RESTART	REstart or STop Antithrombotics Randomised Trial
SAE	Serious adverse event
SAH	Subarachnoid haemorrhage
SDH	Subdural haemorrhage
SAR	Serious adverse reaction
SOP	Standard operating procedure
SPC	Summary of product characteristics
SUSAR	Suspected unexpected serious adverse reaction
TCC	Trial coordinating centre
TIA	Transient ischaemic attack
TMF	Trial master file

UAR Unexpected adverse reaction

VKA Vitamin K antagonist

3 SUMMARIES

3.1 PROFESSIONAL SUMMARY

Primary research question	For adults surviving spontaneous (non-traumatic) symptomatic intracranial haemorrhage with persistent/paroxysmal atrial fibrillation/flutter (AF), does starting full treatment dose oral anticoagulation (OAC) result in a beneficial net reduction of all serious vascular events compared with not starting OAC?
Trial design	Investigator-led, multicentre, randomised, open, assessor-masked, parallel group, clinical trial of investigational medicinal product (CTIMP) prescribing strategies. We plan for a pilot phase, followed by a safety phase.
Objectives	Pilot phase: ~30 hospital sites keep screening logs and recruit at least 60 participants >24 hours after spontaneous symptomatic intracranial haemorrhage with AF and a CHA₂DS₂-VASc score ≥2 to determine the acceptability and feasibility of recruiting the target sample size in a definitive trial in an acceptable timescale. Safety phase: ~60 hospital sites will recruit at least 190 participants to determine whether the risk of recurrent symptomatic intracranial haemorrhage is sufficiently low (non-inferior) to justify a definitive trial.
Eligibility criteria	Inclusion: Spontaneous symptomatic intracranial haemorrhage, AF and a CHA₂DS₂-VASc score ≥2. Exclusion: Patient age <18 years. Symptomatic intracranial haemorrhage within the last 24 hours. Brain imaging that first diagnosed the intracranial haemorrhage is not available. Intracranial haemorrhage is exclusively due to trauma or haemorrhagic transformation of ischaemic stroke. Patient or their doctor is certain about whether to start full dose OAC. Intention to implement the allocated treatment strategy for <1 year. Patient is not registered with a general practitioner. Patient is pregnant, breastfeeding, or of childbearing age and not taking contraception. Lactose intolerance. Patient and carer unable to understand spoken or written English. Previously randomised in SoSTART. Brain magnetic resonance imaging (MRI) sub-study: MRI must be done after intracranial haemorrhage but before randomisation. Sub-study participants must not have contraindications to MRI.
Setting	Recruitment in secondary care (inpatient and outpatient services in stroke, general internal medicine, medicine of the elderly, cardiology, neurology and neurosurgery) with follow-up in primary and secondary care.
Randomisation	Central, web-based randomisation, with 1:1 allocation of intervention: comparator, using a minimisation algorithm.

Intervention	Start long-term (≥ 1 year) full treatment dose OAC (either a non-vitamin K antagonist direct oral anticoagulant [DOAC] or vitamin K antagonist if a DOAC cannot be used), chosen by the patient's physician before randomisation.
Comparators	Do not start OAC (standard clinical practice without OAC may include antiplatelet drug(s) or no antithrombotic drugs).
Outcome measures	Pilot phase: The proportions of eligible patients who are recruited, unsuitable, or decline to participate; the acceptability of the trial protocol to investigators and patients; and the rate of recruitment per site. Safety phase: Primary outcome: Recurrent, symptomatic, spontaneous intracranial haemorrhage. Exploratory outcomes: All symptomatic serious vascular events (i.e. major adverse cardiac or cerebrovascular events [MACCE]) including non-fatal stroke and spontaneous subdural haemorrhage, non-fatal myocardial infarction, vascular death, sudden death, or death of unknown cause. Individual symptomatic vascular events. Individual types of fatal events. Dependence according to the modified Rankin Scale.
Follow up	At least one year after randomisation, using annual questionnaires to participants and their GPs, including review of any medical records and brain imaging relating to outcomes.
Sample size	We plan to recruit at least 60 participants in a pilot phase and at least 190 participants in a safety phase (12% equivalence margin in the outcome of symptomatic intracranial haemorrhage, 1-sided $p=0.025$ and power 90%).

3.2 PLAIN ENGLISH SUMMARY[†]

Background to the research

- Bleeding within the skull, also known as brain haemorrhage, affects 3 million people in the world each year.
- One in five people who survive brain haemorrhage have an irregular heart rhythm called 'atrial fibrillation', which puts them at risk of stroke and other blood clots.
- Blood-thinning medicines, known as 'anticoagulant' drugs, are used in everyday clinical practice to protect people with atrial fibrillation from developing blood clots. However, these drugs also increase the risk of bleeding.
- 'Randomised controlled trials' showed that anticoagulant drugs benefit people with atrial fibrillation. Randomised controlled trials are the fairest tests of treatments. However, these trials did not include people who had a history of brain haemorrhage.
- Recent published studies have observed what happened to people with brain haemorrhage and atrial fibrillation. These studies found that people who started anticoagulant drugs after brain haemorrhage were less likely to die, less likely to suffer blood clots, and were at similar risk of bleeding in comparison to people who did not start these drugs.
- However, these studies were not reliable because they were not randomised controlled trials.

Aims of the research

- We aim to perform a randomised controlled trial involving people with atrial fibrillation who have had a brain haemorrhage.
- We want to produce reliable information about whether these people benefit from starting anticoagulant drugs.
- These findings would be used to look after people with brain haemorrhage and atrial fibrillation in standard clinical practice in the NHS.

Design and methods used

- A network of hospital doctors, nurses, and other staff will identify people who survive brain haemorrhage and have atrial fibrillation.
- If a patient and their doctor are uncertain about whether to start an anticoagulant drug, they may invite the patient to participate.
- Staff will provide the patient, their legal representative, or their nearest relative with information leaflets about this trial.
- After consent is given, a patient can undergo an additional detailed magnetic scan of their brain, known as MRI, which would not always happen in standard practice. After the MRI scan, each patient will be allocated at random to either starting or not starting an anticoagulant drug. There is a 1 in 2, or 50%, chance of receiving either strategy.
- If the patient is allocated to **start** anticoagulant drugs, the patient's doctor will prescribe their choice of anticoagulant drug for the patient.
- If the patient is allocated to **not start** anticoagulant drugs, the patient's doctor will not prescribe an anticoagulant drug.
- We will follow-up all participants for at least one year to determine whether prescribing an anticoagulant drug improves outcome.

[†] INVOLVE. Make it clear campaign, <http://www.invo.org.uk/makeitclear/>, 2014. Improving the quality of plain English summaries for National Institute for Health Research (NIHR) funded research, 2013.

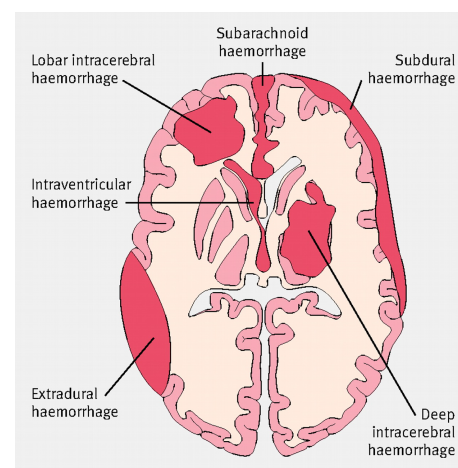
- Participants**
- We started a very similar trial testing ‘antiplatelet’ drugs in the UK in 2013 (called RESTART, www.RESTARTtrial.org), which 537 participants joined.
 - RESTART has involved more than 120 hospitals in the UK, many of which will recruit patients to SoSTART.
 - We will investigate the acceptability of the SoSTART protocol to potential participants in a pilot phase.
- Patient and public involvement**
- The Stroke Association consulted service users, researchers and healthcare professionals to set priorities for research into brain haemorrhage, one of which was prevention strategies (https://www.stroke.org.uk/sites/default/files/haemorrhagic_stroke_research_priorities.pdf).
 - We have consulted the Patient Reference Group of the Research to Understand Stroke due to Haemorrhage (RUSH) programme, who supported the need for this trial and provided feedback on its design (www.ed.ac.uk/clinical-brain-sciences/research/diagnoses-disease-targets/rush/rush-patient-reference-group).
 - A patient is an independent member of our Trial Steering Committee to advise on the conduct and management of the trial.
- Dissemination**
- We will disseminate plain English summaries of the trial’s results to participants who agree to receive these by post, and to the public, using the web and social media.

4 INTRODUCTION

4.1 BACKGROUND

4.1.1 Intracranial haemorrhage has a poor outcome and may recur

Haemorrhagic stroke causes >3 million strokes, half of all stroke deaths and ~42% (47 million) of the disability-adjusted life years lost due to stroke in the world each year.^{1,2} Haemorrhagic stroke includes **intracerebral haemorrhage (ICH)**, **intraventricular haemorrhage (IVH)**, and **subarachnoid haemorrhage (SAH)**, all of which may recur or have a poor outcome.³⁻⁶ Although **subdural haemorrhage (SDH)** is not regarded as a sub-type of haemorrhagic stroke, it can result in similar clinical consequences.^{7,8} These subtypes of intracranial haemorrhage are deemed 'spontaneous' when they usually occur without preceding head injury (although head injury may be a consequence of them). Traumatic intracranial haemorrhage does not provoke the same therapeutic dilemmas as spontaneous intracranial haemorrhage because the risk of recurrence is lower. Extradural haemorrhage occurs after head injury so it is regarded as 'traumatic' in origin.



Intracranial haemorrhage types

4.1.2 Some people with intracranial haemorrhage have atrial fibrillation, which may cause cerebral or systemic embolism

The prevalence of atrial fibrillation (AF) rises with age: in a European population in the early 1990s the overall prevalence of AF was 5.5%, rising from 0.7% at 55–59 years to 17.8% at ≥85 years.⁹ AF is associated with an increased risk of death of any cause, cardiovascular death, major cardiovascular events, ischaemic stroke, ischaemic heart disease, sudden cardiac death, heart failure, chronic kidney disease, and peripheral arterial disease (but not haemorrhagic stroke),¹⁰ largely due to embolism of thrombus from the left atrial appendage. AF affects 14-42% of patients after intracranial haemorrhage (see below).^{8,11-15}

Co-morbidities related to CHA ₂ DS ₂ -VASc ¹⁶	Range of reported prevalence, by type of intracranial haemorrhage (%)			
	ICH ^{11,12}	IVH ¹³	Convexity SAH ¹⁴	SDH ^{8,15}
AF	14-31	42	?	15
Congestive heart failure	8	0	?	?
Hypertension	47-83	67	59	11-60
Age ≥75 years	50	~50	~50	~50
Diabetes mellitus	11-20	0	?	5
Ischaemic stroke/TIA	13-23	0	?	2-7
Cardiovascular disease	10-21	16	?	5-22

Because of these comorbidities, spontaneous intracranial haemorrhage survivors are at risk of ischaemic events that are at least as common as recurrent haemorrhage after ICH,³ which makes doctors consider the use of antithrombotic (anticoagulant or antiplatelet) drugs to prevent ischaemic events.¹⁷

For survivors with AF in particular, the risk of cerebral or systemic embolism is likely to be especially high, and for patients not taking full treatment dose oral anticoagulation (OAC) this risk increases with increasing CHA₂DS₂-VASc score (calculated as below).¹⁶ People with AF and a history of intracranial haemorrhage seem to be at even higher risk of future cerebral or systemic embolism than people with AF without a history of intracranial haemorrhage. Large observational studies have found that the occurrence of intracranial haemorrhage (ICH or subdural haemorrhage) in patients with AF is a risk factor for future cerebral or systemic embolism, independent of CHA₂DS₂-VASc score.¹⁸⁻²¹

CHA ₂ DS ₂ -VASc item	Score*
Congestive heart failure/LV dysfunction	1
Hypertension	1
Age ≥75 years	2
Diabetes mellitus	1
Stroke/TIA/thromboembolism	2
Vascular disease (prior myocardial infarction/peripheral artery disease/aortic plaque)	1
Age 65-74 years	1
Sex category (female)	1

* 1-year risk of cerebral or systemic embolism according to total score: 0 (0.0%), 1 (1.3%), 2 (2.2%), 3 (3.2%), 4 (4.0%), 5 (6.7%), 6 (9.8%), 7 (9.6%), 8 (6.7%), 9 (15.2%)

4.1.3 Primary and secondary prevention of cerebral and systemic embolism in people with AF and no history of intracranial haemorrhage

OAC drugs that have been used the longest include the **vitamin K antagonists (VKAs)**, such as warfarin. VKAs reduce the synthesis of functional vitamin K-dependent factors (factor II, FVII, FIX, FX, as well as protein C and protein S) by interfering with the vitamin K redox cycle. Randomised controlled trials involving people with AF *without a history of intracranial haemorrhage* have shown that full dose VKA OAC drugs (i.e. to achieve target international normalised ratio [INR] 2-3) result in net/overall clinical benefit when used to decrease the risk of cerebral and systemic embolism, despite doubling the risk of intracranial haemorrhage:

- A meta-analysis of eight randomised controlled trials found that full dose VKA OAC was superior to antiplatelet drugs for reducing the risk of all stroke and systemic embolism for patients with AF and no history of ischaemic stroke or transient ischaemic attack (TIA).^{22,23}
- A meta-analysis of two randomised controlled trials found that VKA OAC was superior to antiplatelet drugs for reducing the risk of all serious vascular events for patients with AF and a history of ischaemic stroke or TIA.^{23,24}

However, VKA OAC is complicated by unpredictable and variable dosage requirements, dietary restrictions, interactions with many other drugs, and the need for close therapeutic drug monitoring to keep the INR in the therapeutic range.

Newer **direct oral anticoagulants (DOACs)** inhibit an activated clotting factor, either thrombin/FIIa (dabigatran) or FXa (rivaroxaban, apixaban, edoxaban, and betrixaban), thereby reducing thrombin, fibrin, and clot formation. The pharmacokinetic and pharmacodynamic properties of DOACs are more predictable than those of the VKAs, so routine monitoring of the anticoagulant effect is not required,²⁵ and they have few drug-drug and drug-food interactions. Randomised controlled trials *involving people without a history of intracranial haemorrhage* have shown that DOACs are superior to full dose VKA OAC and superior to antiplatelet drugs, when used to prevent cerebral and systemic embolism in patients with AF:

- A meta-analysis of four randomised controlled trials of four DOACs (dabigatran, rivaroxaban, apixaban, and edoxaban) found that they were collectively superior to warfarin for reduction of cerebral or systemic embolism (without heterogeneity in sub-groups), all-cause mortality and intracranial haemorrhage, although DOACs did cause more gastrointestinal haemorrhages.²⁶
- A meta-analysis of 10 randomised controlled trials found that factor Xa DOACs were superior to full dose VKA OAC for reducing the risk of strokes and systemic embolic events for patients with AF.²⁷ Factor Xa DOACs also reduce the risk of intracranial haemorrhage,²⁸⁻³⁰ but dabigatran and rivaroxaban seem to increase the risk of gastrointestinal bleeding compared to VKAs.³¹ There are no direct randomised comparisons to indicate whether one factor Xa DOAC is superior to the others²⁷ (although an observational study found apixaban was associated with a lower risk of major haemorrhage requiring hospitalisation than other DOACs and warfarin³²). Age does not alter the effect of DOACs compared to warfarin.^{33,34}
- The AVERROES randomised trial found that the DOAC apixaban was superior to aspirin for reducing the risk of stroke or systemic embolism (without increasing the risk of intracranial or extracranial haemorrhage) for patients with AF at increased risk for ischaemic stroke who were not suitable candidates for – or were unwilling to receive – VKA treatment.³⁵

No robust, prospective observational study has shown any benefit from withholding OAC on the basis of a high risk of bleeding³⁶ or a low risk of embolism.³⁷

Other interventions are not superior to full dose OAC. **Antiplatelet drugs** result in a ~20% relative reduction in the risk of stroke for patients in AF, but full dose OAC is substantially more effective than antiplatelet drugs.²³ **Left atrial appendage occlusion** with the WATCHMAN device (followed by prolonged antiplatelet therapy) is an intervention that was non-inferior^{38,39} and possibly superior⁴⁰ to warfarin for the prevention of cerebral and systemic embolism over ~4 years in patients with AF in randomised trials.

Guidelines for the management of AF from The National Institute for Health and Care Excellence (NICE),⁴¹ which concur with other guidelines,^{42,43} recommend:

- Do not offer stroke prevention treatment to people aged <65 years with AF and no risk factors other than their sex (i.e. CHA₂DS₂-VASc score of 0 for men or 1 for women).
- Consider full dose OAC for men with a CHA₂DS₂-VASc score of 1, while taking the bleeding risk into account.
- Offer full dose OAC to people with a CHA₂DS₂-VASc score of ≥2, taking bleeding risk into account.
- Anticoagulate with a DOAC (apixaban, dabigatran etexilate, or rivaroxaban) or a VKA (such as warfarin).

- Discuss options for OAC with the patient and base choice on his or her clinical features and preferences.
- Do not offer aspirin monotherapy solely for stroke prevention to people with AF.
- Do not offer left atrial appendage occlusion as an alternative to OAC unless OAC is contraindicated or not tolerated.

4.1.4 Interventions to reduce serious vascular events for patients with AF and a history of intracranial haemorrhage

75-96% of people with AF after intracranial haemorrhage have at least one other co-morbidity (see 4.1.2 above on page 15) resulting in a CHA₂DS₂-VASc score of ≥ 2 that would justify consideration of full dose OAC based on the CHA₂DS₂-VASc score for predicting embolism.^{16,18,20,44-47} Risk prediction scores have also been developed to predict the absolute risks of bleeding when taking full dose OAC (HAS-BLED⁴⁸ performs well and is simple,⁴⁹ but it has not been compared to Qbleed,⁵⁰ and the recent biomarker-based ABC-bleeding score appears to perform marginally better than HAS-BLED⁵¹). NICE recommends the use of the CHA₂DS₂-VASc and HAS-BLED scores for risk stratification of patients with AF in general,⁴¹ but it is unclear whether full dose OAC should be given to all survivors of intracranial haemorrhage with AF, some survivors according to these risk prediction scores, or none.

4.1.4.1 Full dose oral anticoagulation is associated with better outcome than antiplatelet drugs and no antithrombotic drugs in observational studies

In a systematic review of Medline and EMBASE (updated to 31st December 2016),⁵² we found several observational studies of the outcomes associated with starting vs. not starting OAC after intracranial haemorrhage associated with VKA or DOAC use. Many of these have been small case series (without control groups) involving adults with various indications for full dose OAC, including patients with prosthetic/mechanical heart valve replacements. Given the high risk of thromboembolism with prosthetic heart valves, the dilemma for these patients after intracranial haemorrhage is when – rather than whether – to resume VKA OAC; resumption appears to be associated with overall benefit but the recommended interval has varied from a few days⁵³⁻⁵⁶ to at least one month.⁵⁷ Many of these observational studies have not reported outcomes separately for patients by site of haemorrhage or according to the indication for full dose OAC, which confer different risks of future serious vascular events.^{47,57-79}

Three published studies (and one conference abstract⁸⁰) have focussed on the outcome for adults with AF and intracranial haemorrhage, stratified by antithrombotic drug use after stroke.^{21,44,45} The published studies found that starting full dose OAC (almost exclusively with warfarin) was associated with a lower risk of ischaemic stroke and a higher risk of recurrent intracranial haemorrhage (event rates are summarised in the table below). There were associations between starting warfarin and a lower risk of a composite outcome of ischaemic stroke or ICH²¹ and a composite outcome of any serious vascular event⁴⁴ (i.e. net clinical benefit). Two of these studies also found that restarting full dose OAC was associated with a reduction in the risk of death of any cause,^{44,45} and one of them found an association with better functional outcome (probably mediated by reduction in the risk of recurrent stroke).⁴⁵ One study suggested that a CHA₂DS₂-VASc score of 6 was a threshold about which there was net benefit and below which there was net harm from OAC.²¹ One study found that starting warfarin was associated with lower costs due to hospitalisation.⁸¹

Study	Study design	Antithrombotic drug use after intracranial haemorrhage (n)	Event rate (% per year)		
			Recurrent intra-cranial haemorrhage	Ischaemic stroke	Any serious vascular event
All hospitals in Denmark (Nielsen <i>et al.</i> ⁴⁴)	AF treated with OAC <6 months before intracranial haemorrhage. Inception six weeks after hospital discharge. 1997-2013. Followed 1 year.	1,089 none	8.6	10.4	27.4
		360 antiplatelet	5.3	10.3	23.2
		303 OAC	8.0	5.3	17.1
19 hospitals in Germany (Kuramatsu <i>et al.</i> ⁴⁵)	AF and VKA-associated primary intracerebral haemorrhage. Inception at hospital discharge. 2006-2012. Followed 1 year.	456 none	5.7	14.9	-
		110 warfarin	7.3	5.5	-
Population of Taiwan (Chao <i>et al.</i> ²¹)	AF, CHA ₂ DS ₂ -VASc ≥2, and a past history of intracranial haemorrhage. 1996-2011. Followed for mean 3.3 years.	8,211 none	4.2	5.8	-
		3,552 antiplatelet	5.3	5.2	-
		1,154 warfarin	5.9	3.4	-

Similar therapeutic dilemmas about starting full dose OAC arise after SDH because of the absence of randomised trials and observational studies with dramatic effects.⁸² Similar information from randomised or non-randomised observational studies for people with non-aneurysmal convexity SAH or IVH appear to be lacking despite their ages and comorbidities (see 4.1.2 on page 15). Observational studies of people with AF taking full dose VKA OAC at the time of gastrointestinal haemorrhage have found similar associations between restarting full dose OAC and lower risks of death and thromboembolism (albeit without attempts to account for confounding by indication).⁸³

4.1.4.2 Antiplatelet drugs are not associated with better outcome than no antithrombotic drugs in observational studies

In observational studies of all patients with ICH (i.e. with or without AF), antiplatelet drug use was not associated with worse outcome,⁸⁴⁻⁸⁶ and was associated with a reduction in the risk of all serious vascular events in one study.⁸⁵ Two published observational studies restricted to adults with AF and intracranial haemorrhage have not found associations between antiplatelet drug use and better outcome compared to not using antithrombotic drugs (see table above).^{21,44} Antiplatelet drugs are already being compared to no antithrombotic therapy for patients with AF and a history of ICH in our REstart or STop Antithrombotics Randomised Trial (RESTART, ISRCTN71907627).

4.1.4.3 Left atrial appendage occlusion has been evaluated in uncontrolled case series

Although experts think that left atrial appendage occlusion may be considered for patients with AF after – or at high risk of – bleeding conditions,^{87,88} the WATCHMAN device has

only been used in patients with AF and contraindications to OAC in small observational studies without a comparison group,⁸⁹⁻⁹¹ the safety of the antiplatelet treatment that is required after left atrial appendage occlusion is unknown in people with intracranial haemorrhage (and is the subject of the ongoing RESTART trial),⁹² and left atrial appendage occlusion is not widely available or funded in standard clinical practice. Randomised trials comparing left atrial appendage occlusion versus antithrombotic treatment for people with AF and a past history of intracranial haemorrhage are ongoing (NCT02426944, NCT02830152).

4.2 RATIONALE FOR THE TRIAL

4.2.1 Guidelines, a systematic review, and expert opinions have confirmed therapeutic uncertainty

Recent guidelines and a working party have not found high quality evidence to enable them to make strong recommendations about whether and when to start full dose OAC for people with AF after intracranial haemorrhage.⁹³⁻⁹⁵ Our systematic review (updated to 31st December 2016) did not identify any publications of completed randomised controlled trials comparing full dose OAC to other interventions for people with AF after intracranial haemorrhage.⁵² Several narrative reviews, surveys of experts and debates have been unable to reach agreement on the optimal management of patients in AF with intracranial haemorrhage associated with full dose OAC.⁹⁶⁻¹⁰² A frequently-cited decision analysis using a Markov model compared full dose OAC with warfarin versus no OAC for a hypothetical patient with AF after ICH.¹⁰³ This decision analysis was limited by the use of one hospital-based case series for event rates, an absence of data on the effect of full dose OAC on the risk of recurrent ICH, and the assumptions of Markov models, rendering the authors' conclusions unreliable for use in clinical practice.

4.2.2 Variation in everyday clinical practice

Uncertainty about the effects of full dose OAC after intracranial haemorrhage is likely to have contributed to variation in clinical practice around the world. Although 75-96% of patients with intracranial haemorrhage and AF have a CHA₂DS₂-VASc score of ≥ 2 ,^{18,20,44-47} in the past only 11-33% of these patients have started full dose OAC after intracranial haemorrhage.^{20,44-46,58} In an observational study of 206 ICH survivors who had taken VKA OAC at ICH presentation in everyday clinical practice in 2004-2013 in six European countries, 29 (14%) restarted full dose OAC, 23 (11%) restarted antiplatelet drugs, and one (0.5%) restarted both; there was evidence of significant variation in the proportion of patients restarting OAC between cohorts¹⁰⁴ and this proportion has been lower in Sweden (9-11%),^{46,105} but higher in Denmark (17%),⁴⁴ Germany (24%),⁴⁵ the USA (48%),⁷⁸ and Japan (91%).¹⁰⁶ A survey of neurosurgeons found that the majority relied on intuition or past experience, rather than the available evidence, when making these decisions.¹⁰⁷

4.2.3 Recent evidence that full dose oral anticoagulation appears to be a promising intervention for people with AF after intracranial haemorrhage

Associations observed in recently-published non-randomised studies suggest that starting full dose OAC may be superior to not starting OAC in patients with intracranial haemorrhage (especially ICH) and indications (especially AF) for full dose OAC (see 4.1.4.1 above on page 18). The vast majority of patients with AF and a history of intracranial haemorrhage have used VKA OAC in the epochs of the published

observational studies. In two of these studies, the average durations to starting OAC were 34 days after hospital discharge⁴⁴ and 31 (interquartile range 18-65) days after ICH onset.⁴⁵ Another observational study of ICH survivors with AF found that OAC initiation 7-8 weeks after ICH onset might be the time when there was the largest statistically significant reduction in the 3-year cumulative incidence of ischaemic events in comparison to not using antithrombotic drugs, without any increase in the risk of haemorrhage at any time after ICH.¹⁰⁵ Therefore, it is reasonable to consider starting OAC anytime after the early risk of ICH growth has subsided (24 hours after onset¹⁰⁸), but starting at 1-2 months might maximise the potential benefit of OAC.

In this setting, DOACs are a more promising intervention than VKA OAC because of their collective superiority to warfarin.²⁶ A recent observational study found that starting a VKA OAC or dabigatran after major bleeding in AF was associated with better outcome than not starting OAC, and that starting dabigatran use was associated with a lower risk of recurrent major bleeding.⁷⁸ The factor Xa DOACs also appear promising based on their overall effects²⁷ and their apparent reduction in the risk and severity of intracranial haemorrhage.²⁸ Of all the factor Xa DOACs, apixaban appears the most promising because its beneficial and adverse effects seem not to vary with patient age,³³ it was superior to aspirin for patients with AF who were not suitable candidates for – or were unwilling to receive – VKA OAC,³⁵ and it was associated with the lowest risk of bleeding in observational analyses of all DOACs compared to warfarin.³²

4.2.4 Unknown influence of radiographic biomarkers of cerebral small vessel disease

Brain microbleeds on MRI are radiographic biomarkers of cerebral small vessel diseases¹⁰⁹ that are associated with VKA use in the general population¹¹⁰ and at the time of ICH.¹¹¹ Use of VKA OAC, but perhaps not DOAC, seems to be associated with an increase in the number of brain microbleeds on repeat MRI after one year.^{112,113} Brain microbleeds are risk factors for recurrent ICH in people with ICH,¹¹⁴ but they are also risk factors for ischaemic stroke.¹¹⁵ Uncertainty remains about whether these biomarkers might modify the effects of full dose OAC on recurrent intracranial haemorrhage or ischaemic stroke,¹¹⁶ particularly when cerebral amyloid angiopathy is the suspected small vessel disease underlying lobar ICH,^{103,117} although OAC use after ICH did not appear to differ between lobar and non-lobar ICH.⁴⁵ If there is heterogeneity of the effects of full dose OAC after intracranial haemorrhage by brain microbleed presence or number, this might affect decisions in clinical practice. Careful sub-group analysis of randomised controlled trials are the best way of investigating heterogeneity/modification of treatment effects.¹¹⁸

4.2.5 Randomised trials of starting versus not starting full dose oral anticoagulation for people with AF after intracranial haemorrhage are needed

The magnitude and precision of the observed associations between starting full dose OAC and good outcome in observational studies are not sufficiently ‘dramatic’ by the Oxford Centre for Evidence Based Medicine’s criteria¹¹⁹ to rule out the possibility of selection bias and confounding by indication accounting for the observed associations.

Therefore, the next step to improve the outcome for patients with AF who survive intracranial haemorrhage is to seek definitive proof of the overall benefit of full dose OAC in randomised controlled trials,¹²⁰ and to investigate heterogeneity of treatment effect in sub-groups (e.g. by brain MRI biomarkers).¹¹⁸

4.2.5.1 Ongoing and planned randomised controlled trials

A search of ClinicalTrials.gov on 14 November 2016, using the search terms, “(haemorrhage OR hemorrhage) AND (intracranial OR intracerebral OR subdural OR subarachnoid) AND (anticoagulant OR anticoagulation OR warfarin OR coumadin OR NOAC OR DOAC) | Interventional Studies” yielded 70 records of studies that might be addressing the same therapeutic dilemma as SoSTART. The search revealed one ongoing randomised trial – Apixaban versus Antiplatelet drugs or no antithrombotic drugs after anticoagulation-associated intraCerebral HaEmorrhage in patients with Atrial Fibrillation (APACHE-AF, NTR4526, NCT02565693) – that included its first recruit in January 2015.¹²¹ A similar randomised trial – New Anticoagulants for Stroke Prevention in Patients with Atrial Fibrillation and Previous Intracerebral Hemorrhage (NASPAF-ICH) – began recruitment in 2017 (NCT02998905). The characteristics of these randomised trials and SoSTART are summarised in the table below.

	APACHE-AF	NASPAF-ICH	SoSTART
Participants	n=100, ≥18 years old, with persistent/paroxysmal AF and CHA ₂ DS ₂ VASc ≥3, 7-90 days after OAC-associated ICH/IVH, mRS ≤4.	n=100, with AF and CHADS ₂ ≥2, ≥14 days after intracerebral haemorrhage, no previous lobar ICH, BP <160/100mmHg, mRS ≤4	n≥800, ≥18 years old, with AF and CHA ₂ DS ₂ VASc ≥2, >24 hours after intracranial haemorrhage
Intervention	Apixaban	Apixaban, rivaroxaban, or dabigatran	Any DOAC or VKA
Comparator	Antiplatelet drug(s) or no antithrombotic drug	Aspirin	Do not start oral anticoagulation (antiplatelet drug(s) or no antithrombotic drug)
Treatment allocation ratio	1:1	2:1	1:1
Masking	Outcome assessment only	Outcome assessment only	Outcome assessment only
Primary outcome(s)	Vascular death or non-fatal stroke	Feasibility (recruitment rate), efficacy (ischaemic stroke or recurrent ICH), and safety (major bleeding)	All symptomatic serious vascular events
Timepoint	≥12 months	≥6 months	≥12 months

4.2.6 Start or STop Anticoagulants Randomised Trial (SoSTART) after spontaneous symptomatic intracranial haemorrhage

4.2.6.1 The need for SoSTART

SoSTART is needed because: (1) there is uncertainty about the net benefit of starting full dose OAC for people with AF after any type of spontaneous intracranial haemorrhage, whether or not OAC was used beforehand, (2) variation in clinical practice has accompanied this uncertainty, (3) recent non-randomised observational studies suggest considerable benefit from starting OAC for people with AF after any type of spontaneous intracranial haemorrhage, (4) DOACs appear to be a safer alternative to VKA OAC for this

indication because of the lower risk of intracranial haemorrhage with their use, and (5) similar ongoing or planned randomised trials will not be large enough to detect net benefit and do not include all sub-types of intracranial haemorrhage.

4.2.6.2 The design of SoSTART

In the **pilot phase** of SoSTART involving ~30 hospital sites, research staff will provide fully anonymised screening logs to determine the acceptability and feasibility of recruiting the target sample size in a definitive trial in an acceptable timescale, and recruit at least 60 eligible participants. In the **safety phase** of SoSTART ~60 hospital sites will recruit at least 190 participants to determine whether the risk of recurrent symptomatic intracranial haemorrhage is sufficiently low to justify proceeding to a definitive trial.

4.2.6.3 SoSTART's feasibility

In May 2015, we used Survey Monkey to email an online questionnaire to 289 physicians on the delegation logs at 111 hospital sites participating in the RESTART collaboration, followed by three email reminders to non-responders. Of 289 invitees, one had opted out of Survey Monkey, 17 emails bounced, 109 (38%) did not respond, and 15 partially completed the survey, leaving 147 (51%) complete responses. 95% of respondents were uncertain whether to start full dose OAC in at least some patients with indications for OAC after intracranial haemorrhage. 88% of physicians would randomise patients with AF and a CHA₂DS₂-VASc score ≥ 2 . One quarter of respondents who would randomise patients in such a trial would prefer to use adjusted dose warfarin and the rest would use a DOAC (of whom 50% would use apixaban and 29% would use any new OAC). For the design of a randomised controlled trial, respondents' preferred comparator was not to use any antithrombotic drugs for patients with lone AF (73%), but to use antiplatelet drug(s) for patients with a past history of vaso-occlusive disease (89%). 82% of respondents were just as, or more, keen to participate in SoSTART than RESTART. The evidence in this survey of uncertainty involving all sub-types of intracranial haemorrhage has influenced the design of SoSTART.

We assessed feasibility of attaining our projected sample size in a prospective, community-based inception cohort study of incident stroke due to ICH in the ~850,000 population of NHS Lothian,¹²² and an ICH trial simulator based on this population (www.dcn.ed.ac.uk/ichsimulator/).¹²³ Of 448 adults incident with ICH in NHS Lothian between 1 June 2010 and 31 May 2013, 213 died within 100 days, 181 did not have AF, four adults with AF had CHA₂DS₂-VASc score < 2 , and eight had dementia, leaving 38 (8%) people with ICH and AF who were potentially eligible for SoSTART. In RESTART, we found that one in 12 potentially eligible patients was recruited. So we will initially establish the feasibility of recruiting two patients per site over 18 months to SoSTART at 30 sites within the collaboration of 120 hospital sites supporting our RESTART trial (www.RESTARTtrial.org) to attain a target sample size of at least 60 in a pilot phase.

4.2.6.4 SoSTART's impact

We hope that SoSTART will establish the safety of starting full dose OAC for patients with AF after intracranial haemorrhage, thereby informing guidelines that hitherto have been unable to make strong recommendations about whether to start or not start full dose OAC.⁹³⁻⁹⁵ Resolving this therapeutic uncertainty would be beneficial now given the rising incidence of OAC-associated intracranial haemorrhage,¹²⁴⁻¹²⁶ and in the future given the ageing of the world's population and the rising prevalence of AF with age.⁹

5 TRIAL OBJECTIVES

5.1 PRIMARY OBJECTIVE

SoSTART aims to study at least 190 people aged ≥ 18 years, >24 hours after onset of spontaneous symptomatic intracranial haemorrhage, with AF and CHA₂DS₂VASc score ≥ 2 to determine the safety of starting full treatment dose OAC compared to not starting OAC for the prevention of the primary outcome of recurrent symptomatic intracranial haemorrhage over ≥ 1 year after randomisation.

5.2 SECONDARY OBJECTIVES

- A **pilot phase**¹²⁷⁻¹²⁹ at ~ 30 hospital sites aims to determine:
 - The acceptability of the trial protocol to investigators and patients
 - The proportions of eligible patients who are recruited, unsuitable, or decline to participate
 - The rate of recruiting at least 60 participants to determine the feasibility of recruiting the target sample size in a definitive trial in an acceptable timescale
- A **safety phase** at ~ 60 hospital sites aims to determine:
 - Whether the risk of recurrent symptomatic intracranial haemorrhage is sufficiently low (non-inferior) to justify proceeding to a definitive trial
 - Whether biomarkers of cerebral small vessel disease (e.g. leukoaraiosis on CT or brain microbleeds on MRI) modify the effects of full dose OAC

6 TRIAL DESIGN

SoSTART is an investigator-led, multicentre, randomised, open, assessor-masked, 1:1 parallel group superiority clinical trial of investigational medicinal product (CTIMP) prescribing strategies that are used in standard care.

6.1 PILOT AND SAFETY PHASES

SoSTART aims to recruit at least 190 participants, starting with a pilot phase over ~ 12 months at ~ 30 sites in the United Kingdom. The recruitment rate in the pilot phase will determine the feasibility of recruiting the remaining participants in the safety phase and a definitive trial.^{127,129} The safety phase will use a non-inferiority design to test whether starting OAC is not less safe than not starting OAC.

6.2 BRAIN MRI SUB-STUDY

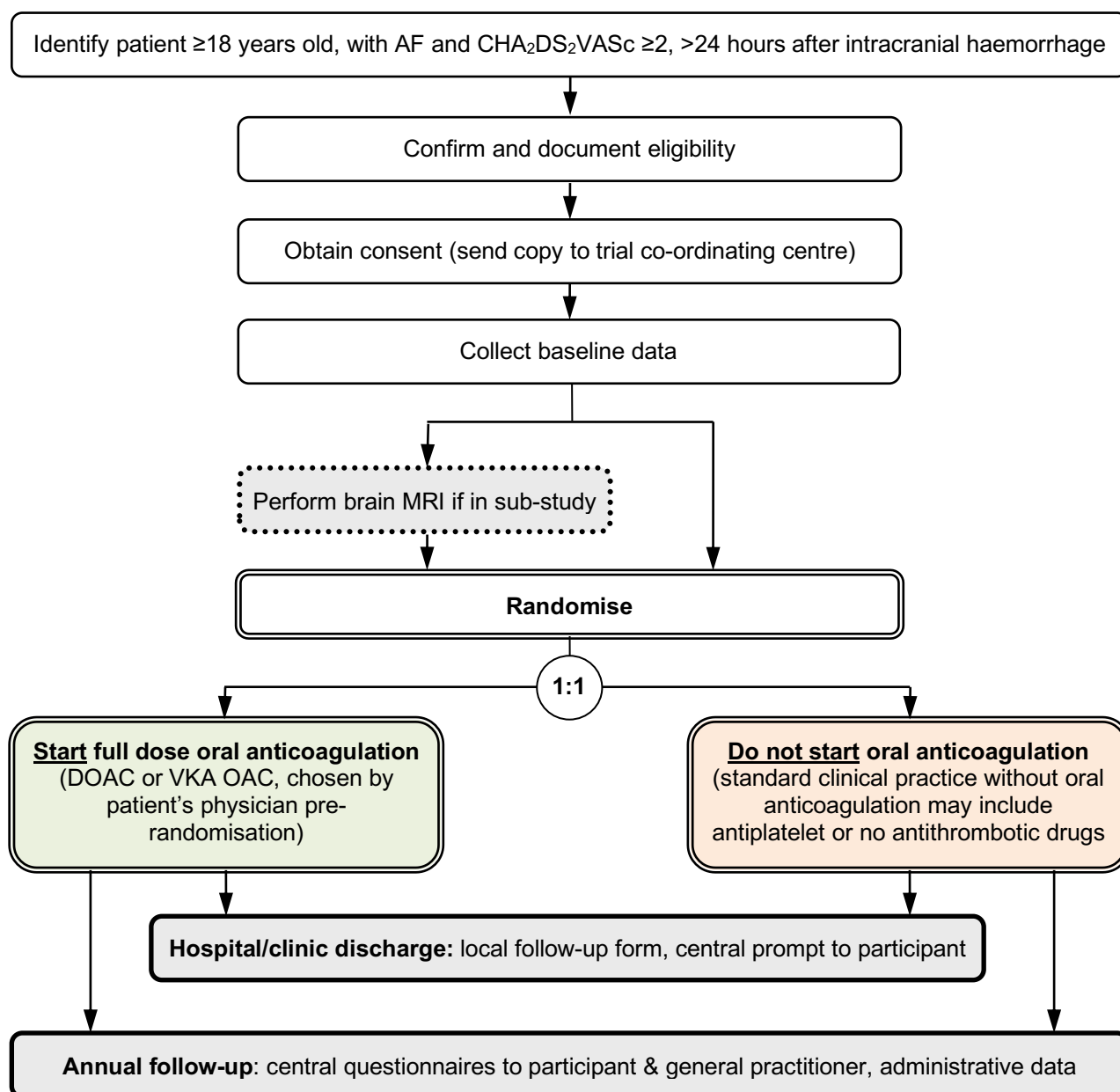
We will perform a brain MRI sub-study in the pilot phase and safety phase to test for modification of the effects of full dose OAC by radiographic biomarkers of small vessel disease (principally, the presence of brain microbleeds).¹¹¹

6.3 TRIAL TIMELINE

	Baseline		Treatment and follow-up		
Assessment	Before randomisation	Randomisation	Discharge	30 days	1 year
Local (hospital site)					
Screen for eligible patients	•				
Check diagnostic imaging is available	•				
Confirm eligibility	•				
Give PILs to patient and/or PLR	•				
Obtain consent and contacts (pCRF)	•				
Complete consent eCRF, record trial ID	•				
Email consent to TCC NHS account	•				
Book and obtain MRI (if in sub-study)	•				
Collect data for randomisation (pCRF)	•				
Submit randomisation form (eCRF)		•			
Record randomisation and treatment allocation in medical records		•			
Prescribe allocated treatment					
Send diagnostic imaging ± MRI to TCC		•			
Submit discharge form (eCRF)			•		
Inform GP about drugs on discharge			•		
Ad hoc reports of AEs and outcomes					
Central (TCC)					
Notify site of treatment allocation		•			
Send GP consent & treatment allocation		•			
Notify participant of treatment allocation			•		
Confirm receipt of brain imaging ± MRI				•	
Confirm receipt of discharge form				•	
GP questionnaire ± reminder (pCRF)					•
Patient questionnaire ± reminder (pCRF)					•

eCRF = electronic case report form, pCRF = paper case report form

6.4 TRIAL FLOWCHART



7 TRIAL SETTING

Recruitment in secondary care (inpatient and outpatient services in stroke, neurology, neurosurgery, general internal medicine, and medicine for the elderly) with follow-up in primary and secondary care.

8 TRIAL POPULATION

8.1 NUMBER OF PARTICIPANTS

We aim to enrol at least 190 participants in the safety phase of SoSTART, beginning with at least 60 participants in the pilot phase in the UK, which we will keep 'internal'^{127,129} allowing the possibility of running seamlessly into the safety phase. We aim to enrol ~60% of participants into the brain MRI study.

8.2 INCLUSION CRITERIA

- Patient age ≥ 18 years
- Symptomatic intracranial haemorrhage (i.e. intracerebral haemorrhage, non-aneurysmal subarachnoid haemorrhage, intraventricular haemorrhage, or subdural haemorrhage)
 - Not attributable to a known underlying intracranial aneurysm, arteriovenous malformation, cerebral cavernous malformation, dural arteriovenous fistula, intracranial venous thrombosis
 - Not attributable to known head injury, based on:
 - a history from the patient/witness of spontaneous symptom onset without preceding head trauma (head trauma occurring after symptom onset is permissible)
 - brain imaging appearances consistent with spontaneous intracranial haemorrhage (which may be accompanied by the brain/bone/soft tissue appearances of trauma occurring subsequently)
- Atrial fibrillation/flutter (persistent or paroxysmal) with a CHA₂DS₂-VASc score ≥ 2
- If included in the brain magnetic resonance imaging (MRI) sub-study, the scan must be done after symptomatic intracranial haemorrhage and before randomisation

8.3 EXCLUSION CRITERIA

- Symptomatic intracranial haemorrhage within the last 24 hours (when the risk of haemorrhage expansion/growth is greatest)
- Symptomatic intracranial haemorrhage is exclusively due to trauma or haemorrhagic transformation of ischaemic stroke
- Prosthetic mechanical heart valve or severe (haemodynamically significant) native valve disease
- Left atrial appendage occlusion for prevention of systemic embolism in AF done in the past, or intended to be performed
- Intention to start antiplatelet drug(s) if randomised to start full dose OAC
- Intention to start OAC or parenteral anticoagulation
- Intention to implement the allocated treatment strategy for <1 year

- Patient or their doctor is certain about whether to start full dose OAC
- Brain imaging that first diagnosed the intracranial haemorrhage is not available
- Patient is not registered with a general practitioner
- Patient is pregnant, breastfeeding, or of childbearing age and not taking contraception. All women of child-bearing potential must have a highly sensitive urine pregnancy test to confirm that they are not pregnant before randomisation
- Patient and carer unable to understand spoken or written English
- Lactose intolerance
- Contraindications to any of the IMPs, other than recent intracranial haemorrhage (see section 10)
- Contraindication to MRI (brain MRI sub-study)
- Life expectancy less than one year
- Previously randomised in SoSTART

8.4 CO-ENROLMENT

Patients in a phase 1 or first-time-into-human study cannot be co-enrolled in SoSTART. Inclusion in another randomised trial or observational study does not preclude participation in SoSTART as long as: participants are not overburdened; their inclusion would be unlikely to confound SoSTART's results or complicate attribution of serious adverse events and outcomes; the protocol of the other study does not preclude co-enrolment in SoSTART; and co-enrolment has been agreed with the Chief Investigators of all studies involved in co-enrolment. Research staff should obtain permission to enrol patients who are participants in other CTIMPs from the SoSTART Chief Investigator via the Trial Coordinating Centre (TCC) or by NHS email. A record of participants who are known to have been co-enrolled in other studies will be maintained by the TCC.

Arrangements for co-enrolment with another CTIMP will be bound by a written agreement between the Chief Investigator and Co-Sponsors of both/all CTIMPs implicated, informed by the completion of the Sponsor's CTIMP-CTIMP Co-enrolment Checklist (POL008-F01). This agreement will include

- special safety reporting measures (if required);
- a minimum wash-out period between last dose in one study and first dose in another;
- a statement to indicate that the chairs of the TSC/DMC from each study and statisticians from each study have no objections to the proposals for co-enrolment;
- a statement that arrangements for attribution of liability for co-enrolled participants have been put in place and agreed between the sponsors of both/all CTIMPs implicated.

9 PARTICIPANT SELECTION AND ENROLMENT

9.1 IDENTIFYING PARTICIPANTS

9.1.1 Research staff

Research staff who may identify potential participants for SoSTART should be part of, or affiliated to, the clinical team responsible for the patient's care, including **doctors** (stroke physicians, geriatricians, neurologists, and neurosurgeons) and **clinical or research staff**.

Research staff will be based at hospital sites where local resources would facilitate patient identification and eligibility assessment. Potentially eligible patients may be identified in the everyday clinical practice of research staff, or referred to them for assessment of eligibility having been identified elsewhere (e.g. by clinical staff who are not research staff in secondary or primary care). These members of the local research team will undergo standardised training on trial-related procedures. At local sites, in this protocol we refer to:

- **principal investigators** (the lead doctor at a research site)
- **co-investigators** (doctors with delegated responsibilities for e.g. eligibility assessment and adverse event reporting)
- **local coordinator** (the lead member of research staff at a research site)
- **local research staff** (who support the principal investigator and co-investigators)

9.1.2 Timing

These research staff should identify patients with a history of symptomatic intracranial haemorrhage and **at a time when uncertainty arises** about whether to start full treatment dose OAC. There is **no specific time window** for identifying participants, so they may be recruited at least 24 hours after symptom onset, during their hospital admission for the qualifying intracranial haemorrhage or at a later stage in an outpatient clinic. Clinical or research staff with delegated responsibility should approach eligible patients, or their personal / legal / professional representative (if the patient lacks mental capacity to consent for themselves), to ascertain their interest in participating in SoSTART and permission to pass their details on to any other research staff involved.

9.2 CONSENTING PARTICIPANTS

9.2.1 Responsibilities

The Principal Investigator (PI), or co-investigator is responsible for confirming eligibility, ensuring informed consent is obtained and that the informed consent form (ICF) is completed, signed and dated by all parties and sent to the SoSTART trial office before randomisation and any protocol-specific procedures are carried out. Local research staff should follow the laws that govern consent procedures in their jurisdiction, and in particular those governing incapacitated adults and their involvement in research.

9.2.2 Process

Patients or their representatives must receive adequate oral and written information, so research staff should provide the appropriate Participant Information Leaflets (PIL) and ICF. The oral explanation to the patient should be performed by the PI, co-investigator or trained research staff member, and must cover all the elements specified in the PILs and ICF. The patient must be given every opportunity to clarify any points they do not understand and, if necessary, ask for more information. The patient should be given as much time as they need to reflect on the information and decide about their participation. The patient will be told that they may withdraw their consent to participate at any time without their medical care being affected. The patient or their representative will provide contact details (address and telephone numbers) for the patient, relatives or carers who are willing to provide information about the patient's health during follow-up). The patient or their representative will be given as much time as they need to consider giving consent for the trial.

9.2.3 Documentation

The PI, co-investigator or research staff member and the participant should sign and date the ICF to confirm that consent has been obtained. Written informed consent should always be sought from the participant where possible. If this is not possible because the participant cannot write, the randomising member of the research team can gain witnessed verbal consent. The participant or their personal legal representative should receive a copy of the ICF, a copy should be filed in the patient's medical records, a copy should be sent to the TCC and the original ICF should be filed in the investigator site file (ISF) along with the randomisation form. Confirmation of eligibility, consent, and the version of the PILs used should be recorded in the participant's paper and/or electronic medical records for any future source data verification, including the date of consent, that the participant received the PILs, who obtained consent and signed and dated confirmation that the patient was eligible for enrolment. Obtaining consent should be recorded in the trial database, which results in the allocation of a unique trial identification number, regardless of whether the patient is subsequently randomised.

9.2.4 Principal Investigator responsibilities to the participant

The participant should receive a copy of the current PILs, a copy of the completed ICF and a copy of the contact form.

9.2.5 Principal Investigator responsibilities to the trial

A copy of the ICF should be sent to the TCC (e.g. via secure, encrypted NHSmail), ideally before randomisation or soon afterwards, as this has to be provided to the GP along with the notification that the patient has been recruited into the trial (see section 19 below on page 52). Local research staff will be prompted by email from the TCC to provide the ICF if it has not been received.

9.2.6 Consenting patients who lack capacity to consent themselves

If a patient lacks capacity to consent for themselves then a personal / legal / professional representative may provide consent on the patient's behalf. The table below specifies the hierarchy that should be applied according to location in the UK.

Hierarchy of informed consent for an adult with mental incapacity

<i>England, Wales and Northern Ireland</i>	<i>Scotland</i>
1. Personal legal representative A person not connected with the conduct of the trial who is: (a) suitable to act as the legal representative by virtue of their relationship with the adult <i>and</i> (b) available and willing to do so. 2. Nominated legal representative Doctor primarily responsible for the incapacitated adult's treatment or a person nominated by their healthcare provider.	1. Personal legal representative Any guardian or welfare attorney who has power to consent to the adult's participation in research. If there is no such person, then ... 2. Nearest relative Adult's nearest relative as defined in section 254 of the Mental Health (Care and Treatment) (Scotland) Act 2003. If it is not reasonably practicable to contact 1 or 2, then... 3. Doctor primarily responsible for the incapacitated adult's treatment or a person nominated by their healthcare provider.

9.2.7 Re-consenting patients who regain capacity

If a participant regains capacity during any hospital stay, the participant should be informed about their enrolment in the trial and fully informed consent should be obtained from the participant. If the participant regains mental capacity after hospital discharge but during follow-up we will not attempt to re-consent at that stage because the nature of the follow up will make it impractical to know whether the participant has regained capacity. The participant who does regain capacity may withdraw their consent to participate at any time, by calling the trial office, without their medical care being affected.

9.3 SCREENING FOR ELIGIBILITY

SoSTART investigators may use the following strategies to achieve adequate participant enrolment, including the following sources of information about patients looked after at their own hospital organisation:

- Admissions to inpatient services for stroke, neurology, neurosurgery, emergency medicine and medicine for the elderly
- Appointments at outpatient follow-up clinics
- Audit databases
- Hospital admission/discharge coding data
- Services' own bespoke databases

During the pilot phase, sites will be required to keep and provide anonymised screening log data to the TCC on all potentially eligible patients at each site, which patients were recruited to SoSTART, and reasons why the others were not recruited. Patient initials will be recorded as a safeguard against duplicate entries, but personally identifying information

will not be recorded. These data will help confirm the suitability of the eligibility criteria, understand recruitment,¹³⁰ and inform the feasibility of conducting a definitive trial.^{127,129}

9.4 INELIGIBLE AND NON-RANDOMISED PARTICIPANTS

Patients are not regarded as being included in SoSTART for the purposes of safety reporting until they have been randomised. SoSTART is not responsible for potentially eligible patients who, ultimately, are not randomised in SoSTART.

10 INVESTIGATIONAL MEDICINAL PRODUCTS

10.1 INTERVENTIONS

OAC listed in the British National Formulary belong to the following classes: factor Xa inhibitors, direct thrombin inhibitors, and VKA OAC. OAC will be used at full treatment dose (i.e. the treatment dose used for prevention of cerebral/systemic embolism in AF, and not 'prophylactic' low dose used for prevention of venous thromboembolism in immobile patients).

10.1.1 Intervention identification

- Factor Xa inhibitors:
 - Apixaban tablet (Eliquis, Bristol-Myers Squibb Pharmaceuticals Ltd)
 - Rivaroxaban tablet (Xarelto, Bayer Plc)
 - Edoxaban tablet (Lixiana, Daiichi Sankyo UK Ltd)
- Direct thrombin inhibitor:
 - Dabigatran etexilate capsule (Pradaxa, Boehringer Ingelheim Ltd)
- Vitamin K antagonists:
 - Warfarin sodium tablet or oral suspension (Warfarin/Marevan/Coumadin, too many different manufacturers to list)
 - Acenocoumarol tablet (Sinthrome, Merus Labs Luxco S.a R.L.)
 - Phenindione tablet (Phenindione, Concordia International)

10.1.2 Intervention manufacturers

No specific manufacturer will be required. Participants will receive the formulations available on their physicians' local formularies or at their hospital/community pharmacies.

10.1.3 Marketing authorisation holder

Not applicable.

10.1.4 Labelling and packaging

Participants will receive the drug as it is dispensed by their pharmacist. No specific labelling or packaging will be used for this trial.

10.1.5 Storage

No specific storage instructions will be given in addition to what is usually advised in the instructions given to patients when OAC drugs are dispensed.

10.1.6 Summary of Product Characteristics or Investigators Brochure

Representative Summaries of Product Characteristics (SPCs) will be used and provided for SoSTART investigators in a separate document that should be kept in the ISF, and will be emailed to sites whenever SPCs are updated.

10.2 COMPARATORS

The comparator will be standard clinical practice without OAC. In standard clinical practice, participants' physicians may choose not to prescribe any antithrombotic drugs, or they may choose to prescribe an antiplatelet drug listed in the British National Formulary if the patient has an indication for antiplatelet drug use. At randomisation, physicians will specify the intended non-investigational medicinal product(s) that they would use if the participant was randomised to not start OAC, to inform the minimisation algorithm. SoSTART will not enforce the use of a specific comparator.

10.2.1 Intervention manufacturers

No specific manufacturer will be used. Participants will receive the formulations available on their physicians' local formularies or at their hospital/community pharmacies.

10.2.2 Marketing authorisation holder

Not applicable.

10.2.3 Labelling and packaging

Participants will receive the drug as it is dispensed by their pharmacist. No specific labelling or packaging will be used for this trial.

10.2.4 Storage

No specific storage instructions will be given in addition to what is usually advised in the instructions given to patients when these drugs are dispensed.

10.2.5 Summary of Product Characteristics or Investigators Brochure

Representative SPCs are provided in a separate document.

10.3 DOSING REGIME

IMPs will be prescribed at the dosage recommended for their indication as specified in the SPCs (see SPCs listed in a separate document).

10.4 DOSE CHANGES

Adjustment of OAC dosage may be required according to renal function, age, body weight, and concomitant medications, as specified in the SPCs (see SPCs listed in a separate document).

10.5 PARTICIPANT ADHERENCE

Regardless of treatment allocation, we will record adherence by requiring investigators to document the antithrombotic drug(s) prescribed after randomisation on the hospital/clinic discharge form. Monitoring of adherence for the remainder of the study period will rely on annual questionnaires completed by participants (or their carer) and participants' GPs, which have proven to be sufficient in another long-term secondary prevention trial (RESTART). To increase the likelihood that participants will receive the allocated intervention, we will:

- Encourage the randomising clinician to emphasise the importance of adhering to the allocated treatment policy to the participant and their GP.
- Write to the GP shortly after enrolment to alert them to the participant's inclusion in the trial and ask them to inform the TCC of any changes to the participant's antithrombotic drug use.
- Write to the participant at home (when the clinic/hospital discharge form is received) reminding them of the purpose of SoSTART and the importance of adhering to the allocated treatment. Participants can use the TCC's contact details to feedback any concerns, which we would respond to.

10.6 OVERDOSE

There is a small risk that the participant, or someone close to them, may intentionally or accidentally take an overdose of antithrombotic drugs, but since SoSTART is an open trial the ingestion of an overdose of antithrombotic drugs could be recognised and managed conventionally.

10.7 OTHER MEDICATIONS

10.7.1 Non-Investigational Medicinal Products

Physicians may intend to prescribe antiplatelet drugs for patients eligible for SoSTART in standard clinical practice who are not allocated to start OAC. Therefore, SoSTART will collect information about these non-investigational medicinal products at randomisation, discharge, and during follow-up.

10.7.2 Permitted medications

Participants' usual medications are permitted. When we inform each participant's GP of the patient's participation in SoSTART, we will make reference to the *National Clinical Guideline for Stroke*, which recommends antihypertensive drugs to consistently achieve a clinic systolic blood pressure below 130mmHg for secondary prevention after stroke.¹³¹ SoSTART cannot be prescriptive about the use of such treatment, in case the trial contradicts a participant's clinician who has not recommended antihypertensive drugs for other clinical reasons. For findings to be generalisable to clinical practice, we will not aim

to optimise blood pressure control as a co-intervention in this trial. However, we do intend to collect data during follow-up on antihypertensive drug prescriptions and blood pressure measurements made by GPs and nurses in the practice electronic patient record.

10.7.3 Prohibited Medications

Other than the use of parenteral anticoagulation, there are no restrictions on medications that participants may take, other than those that are contraindicated because they may interact with the intervention or comparator, as specified in the SPCs (see SPCs listed in a separate document).

11 OUTCOMES

11.1 PRIMARY OUTCOME

11.1.1 Pilot phase

Rate of participant recruitment per site.

11.1.2 Safety phase

Recurrent symptomatic spontaneous intracranial haemorrhage.

11.2 SECONDARY OUTCOMES

11.2.1 Pilot phase

The proportions of eligible patients who are recruited, unsuitable, or decline to participate. The acceptability of the trial protocol to investigators and patients.

11.2.2 Exploratory outcomes in the safety phase

Symptomatic serious vascular events:

- All symptomatic serious vascular events (i.e. major adverse cardiac or cerebrovascular events [MACCE]) including: non-fatal (i.e. not followed by death within 30 days of onset) myocardial infarction; stroke (i.e. ischaemic, haemorrhagic, unknown sub-type) or spontaneous subdural haemorrhage; or death from a vascular cause (i.e. haemorrhagic or ischaemic events followed by death within 30 days), sudden death, or death of an unknown cause.

Individual symptomatic vascular events:

- Major haemorrhagic events (Bleeding Academic Research Consortium types 3-5)¹³²
 - Recurrent symptomatic spontaneous intracranial haemorrhage
 - Extracranial haemorrhage: Overt bleeding plus haemoglobin drop of 3g/dL or more (provided haemoglobin drop is related to bleed). Any transfusion with overt bleeding. Cardiac tamponade. Bleeding requiring surgical intervention for control (excluding dental/nasal/skin/haemorrhoid). Bleeding requiring intravenous vasoactive agents. Intraocular bleed compromising vision.

CABG-related bleeding (reoperation after closure of sternotomy for the purpose of controlling bleeding). Transfusion of 5U or more whole blood or packed red blood cells within a 48-hour period. Chest tube output 2L or more within a 24-hour period. Probable fatal bleeding; no autopsy or imaging confirmation but clinically suspicious. Definite fatal bleeding; overt bleeding or autopsy or imaging confirmation.

- Symptomatic ischaemic events
 - ischaemic stroke
 - myocardial infarction
 - peripheral arterial occlusion
 - mesenteric ischaemia
 - central retinal arterial occlusion
 - deep vein thrombosis
 - pulmonary embolism
 - cardiac death with symptoms suggestive of myocardial ischaemia (type 3),¹³³ or evidence of arrhythmia
- Revascularisation procedures (carotid, coronary, or peripheral arterial)
- Symptomatic stroke of uncertain sub-type
 - Non-fatal stroke, with brain imaging performed too late to distinguish haemorrhage from infarction
 - Rapidly fatal stroke, but without radiographic or pathological confirmation

Individual types of fatal events:

- Vascular deaths
- Sudden deaths
- Deaths of an unknown cause
- Deaths from any other cause

Annual ratings of participant dependence and quality of life completed by participant, their carer or nominated contact, or healthcare provider (e.g. general practitioner):

- Simplified modified Rankin Scale postal questionnaire^{134,135}
- Structured interview to determine the modified Rankin Scale score with non-responders to the postal questionnaire¹³⁶
- The 5-level EQ-5D version (EQ-5D-5L) of the EuroQol

12 ASSIGNMENT OF INTERVENTIONS

Having confirmed eligibility, confirmed that the brain imaging that diagnosed the intracranial haemorrhage is available, and obtained consent, investigators collect the baseline data necessary to complete a randomisation form. The investigator enters the participant's baseline data into a 24-hour computerised central randomisation service by means of a secure web interface (or a telephone call to the TCC during office hours if the web interface is not operational). After range, consistency and validation checks in the online interface, with on-screen prompts, the randomisation service determines the treatment allocation and conveys it and the participant's unique trial identification number by email. Investigators file the completed randomisation form (including the participant ID and treatment allocation) in the ISF. Investigators must indicate the allocated treatment in the participant's paper and/or electronic medical records. Following randomisation, the TCC generates and sends a letter to inform the participant's GP about the participant's enrolment in the trial and treatment allocation, including a copy of the ICF and the follow-up schedule.

12.1 ALLOCATION SEQUENCE GENERATION

To avoid predictable alternation of treatment allocation (and consequent loss of allocation concealment) the computerised, dynamic minimisation algorithm will randomly allocate the first participant with a probability of 0.5 to one arm of the trial. The randomisation algorithm for each subsequent participant involves adaptive stratification (i.e. minimisation) and allocates them with a probability of 0.8 to the group which minimises differences between the two arms of the trial with respect to variables collected by research staff before randomisation. A separate description of the randomisation system is held by the TCC and ECTU.

12.2 ALLOCATION CONCEALMENT MECHANISM

The web-based, central, computerised randomisation service dynamically allocates treatment according to the minimisation algorithm, without a pre-determined allocation sequence, and only reveals treatment allocation for an individual participant after the local investigator has submitted the randomisation form.

12.3 IMPLEMENTATION OF TREATMENT ALLOCATION

The local researcher will be provided with a participant's treatment allocation by the web interface (or by telephone if the web interface is not operational). The randomisation system automatically generates an email to confirm the treatment allocation. The intervention and comparator in SoSTART will be prescribed by the PI or co-investigator in clinic or on the ward, and dispensed within 24 hours of randomisation. The TCC will send a letter to the participant confirming the treatment allocation as-randomised and information about how to report adverse events to the SoSTART TCC (see section 19 below on page 52).

12.4 MASKING (BLINDING)

Treatment allocation in SoSTART is not masked, and therefore it is open to participants, the clinicians caring for them in secondary and primary care, and local research staff. However, the outcome event adjudication committee will be masked to participants' allocated treatment when assessing clinical and radiographic information about outcome events.

12.5 EMERGENCY UN-MASKING PROCEDURES

Not required, because treatment allocation is open.

13 WITHDRAWAL OF TRIAL PARTICIPANTS

Participants are free to withdraw from the trial at any point or a participant can be withdrawn by an investigator, treating physician or Sponsor. If withdrawal occurs, the primary reason for withdrawal must be documented in the participant's case record form. The participant will have the option of withdrawal from any or all of:

- allocated OAC treatment policy
- participant postal follow-up questionnaires

- participant follow-up questionnaire conducted by telephone
- general practitioner follow-up questionnaires
- follow-up using record linkage

A trial participant should stop OAC treatment if they develop a new adverse event which in the opinion of the investigator jeopardises the safety of the trial participant, or if they become pregnant. Advice from the DMC to stop the trial might lead to early discontinuation. Withdrawn participants will not be replaced.

14 ASSESSMENTS, SOURCE DATA DOCUMENTATION AND DATA COLLECTION

The PI, co-investigators and local researcher staff at each site will collect the local data listed in the schedule of assessments below and the trial timeline (see section 6.3 on page 25). Research staff in the TCC will collect the central data.

14.1 CASE REPORT FORMS

Data will be collected using electronic case report forms (eCRF) via the secure web interface to the trial database. Paper CRFs will be made available to facilitate data complete data collection for lengthy forms (e.g. baseline data on the randomisation form). The data that will be collected by each eCRF are listed below at each time point.

14.2 ELIGIBILITY, CONSENT, AND CONTACT INFORMATION

14.2.1.1 Data collection

After establishing eligibility and completing the consent procedures, local research staff will complete the consent eCRF to indicate: who confirmed eligibility; participant's name, address, and health service identification number; what the patient has consented to; who provided consent; who obtained consent; when consent was obtained; and the contact details (landline, mobile telephone number, and postal address) of the patient and people who the patient confirms agree to be contacted for information on the patient's progress during follow-up should the patient be uncontactable. Entry of these data will generate a unique ID number for the consented participant.

14.3 RANDOMISATION

14.3.1 Clinical

Following consent, local research staff will supply the baseline data required by the randomisation eCRF via the SoSTART website at the time of randomisation. After randomisation, research staff will record the participant's trial number and treatment allocation in the participant's medical records.

14.3.1.1 Data collection

Local investigators will collect these data directly from patients/carers, medical records and brain imaging reports, including discussion with the PI or investigators if required. The

following data will be collected: trial co-enrolment details; participant's co-morbidities, indications for antithrombotic therapy, type of AF (persistent/paroxysmal) and variables needed to predict the risk of thrombo-embolic events (CHA₂DS₂-VASc)¹⁶ and haemorrhagic events (HAS-BLED)⁴⁸ in AF, as these appear to be the simplest scores with the best predictive ability¹⁹; last serum creatinine measurement; two sedentary blood pressure readings; antithrombotic drugs (if any) taken before intracranial haemorrhage; date and type of qualifying intracranial haemorrhage; details of imaging performed; intended choice of intervention and comparator; physician's reasons for uncertainty; patient's functional status; GP's contact details; and health-related quality of life (using the EuroQol Group EQ-5D-5L instrument).

14.3.2 Radiographic

Investigators will be required to verify at the time of randomisation that the brain imaging that diagnosed the qualifying intracranial haemorrhage is available. If a participant gives consent to join the MRI sub-study, at the time of randomisation the researcher will need to confirm that the MRI has been performed. We require gradient recalled echo (GRE) T2*, T1, T2, FLAIR and DWI MRI sequences to be performed, with susceptibility-weighted imaging if it is available. If a participant has already undergone brain MRI as part of their routine clinical care with the required sequences before randomisation, SoSTART will request the MRI and the participant will not need to enter the MRI sub-study. An investigator must send the relevant brain imaging to the TCC in anonymised DICOM format accompanied by an imaging transfer form. Investigators will be reimbursed for providing protocol-compliant brain imaging in accordance with the details outlined in the Clinical Study Site Agreement and the patient is randomised. The radiology department at each site will issue the clinical report of any sub-study MRI.

14.3.2.1 Data collection

Following receipt of brain imaging, central staff will check each imaging study to ensure that it relates to the appropriate participant at the appropriate time, that it is the appropriate modality, and that all the required images and sequences have been sent. After quality assurance and obtaining required imaging, anonymised images will be uploaded to an electronic archive for later rating according to agreed standards.¹⁰⁹ These data will be stored as part of each participant's eCRF in an imaging database.

14.4 DISCHARGE

After a participant is randomised and when they are discharged from the hospital or outpatient clinic, the PI, co-investigators or research staff will complete the SoSTART discharge form. When the TCC receives the discharge form, the participant will be sent a letter from the TCC reminding them of their treatment allocation and the follow-up schedule.

14.4.1.1 Data collection

Local investigators or research staff will submit information via an eCRF about: the responsible consultant at the time of the participant's discharge; the names of all prescribed antithrombotic and antihypertensive drugs; any outcomes that occurred between randomisation and discharge; the participant's discharge destination; and updated contact details for the participant if they have changed since randomisation.

14.5 ANNUAL FOLLOW-UP

14.5.1 General practitioner questionnaire

About four weeks before the anniversary of a participant's randomisation, the TCC will send each participant's GP (or hospital consultant, if a discharge form has not been received) a paper questionnaire and a copy of the signed ICF; if no response is received this will be followed by a telephone reminder after the anniversary of a participant's randomisation. GPs will be reimbursed as recommended by the Primary Care Research Network.

14.5.1.1 Data collection

Respondents will be asked to provide details from their medical records on a paper CRF of: the participant's vital status; the occurrence of primary and secondary outcome events and any hospital admissions since the discharge form was completed; the participant's three most recent BP measurements; a list of current repeat prescriptions (including antithrombotic and antihypertensive drug names); and up-to-date contact details for the participant. These data will be entered into an equivalent eCRF in the trial database by TCC staff.

14.5.2 Participant questionnaire

If the GP indicates that the participant is still alive and the participant has not withdrawn from questionnaire follow-up, the participant will be sent a paper questionnaire (followed by a telephone reminder if no response is received).

14.5.2.1 Data collection

Participants will be asked to provide: details of primary and secondary outcome events since randomisation; hospital admissions since discharge; answers to structured questions that permit calculation of their current modified Rankin Scale score^{134,136} and health-related quality of life (using the EuroQol Group EQ-5D-5L instrument); and a list of their repeat prescriptions (including antithrombotic and antihypertensive drugs). If the participant has incapacity or cannot speak English, a carer or relative may complete and return the forms. Experience in previous trials indicates that failure to complete a postal questionnaire usually indicates a failure of receipt or inadvertent non-completion rather than a wish not to participate further in the trial. If the follow up information cannot be obtained by either postal or telephone questionnaire the local research team will be asked to provide follow up information. These data will be entered into an equivalent eCRF in the trial database by TCC staff.

14.5.3 Ad hoc follow-up

Contact with the TCC is available during office hours for participants, carers, GPs, site investigators and other research staff to report or discuss outcome events, adverse events, and any other issues that are pertinent to participants. Information acquired in this way will be stored electronically or on paper in the trial master file. If the TCC learns of an outcome event from any or all of the sources above, the TCC will collect information required to confirm its occurrence and accurately classify it.

14.5.3.1 Data collection

The TCC will obtain some or all of the following information (as appropriate) about events reported to the TCC by participants, carers, GPs, site investigators and other local research staff at collaborating sites: medical records of emergency assessments by the GP or Emergency Department staff; hospital discharge summary; inpatient medical or nursing records; brain imaging; death certificate; Coroner (or Procurator Fiscal) autopsy report; interview with legal representative / nearest relative (if they agree, and none of the above information can be obtained).

14.6 EVENT ADJUDICATION

Events that are reported to the TCC will be internally reviewed by members of the trial management group. If these events might be primary or secondary outcomes, or SAEs/SUSARs, TCC staff will obtain all relevant clinical, radiographic or pathological information about a reported event. Patient identifiers and information about antithrombotic treatment will be redacted from these documents and internally adjudicated by a physician at the TCC if there is any uncertainty about the type of outcome event. If brain imaging has been performed in relation to a reported event, the TCC will obtain the clinical report on any relevant brain imaging. Redacted information about all events that are reported, and which internal adjudication classifies as definitely or possibly outcome events, will be sent to an independent outcome event adjudicator. Any reported events that are identified during the review process as potential SAEs or SUSARs will be reported to the sponsor as described below.

14.7 LONG-TERM FOLLOW-UP AFTER THE TRIAL PERIOD

Consent will be obtained for long-term follow-up of the trial participants after the trial period ends. This may occur if the trial enters a main phase by record linkage with primary care and hospital discharge and flagging with NHS Digital to determine long-term outcome. These linkages may also contribute to the identification of outcomes during the trial follow-up period.

14.8 DATABASE AND DATA MANAGEMENT

The Edinburgh clinical trials unit will provide the trial database. Access to the databases will be restricted (requiring an individual ID and password to log in). All users will require training before access is granted. The TCC and ECTU will provide support for users during working hours. The database will have a secure audit trail. The database will have back up procedures to minimise potential data loss.

Local research staff will enter baseline, randomisation and hospital/clinic discharge data into the trial database using eCRFs. At randomisation, this will be done via the secure trial website. During follow-up, central research staff will record data provided via ad hoc reports from investigators and participants (or their carers) and annual questionnaires from GPs and participants (or their carers). Data entry errors will be minimised by using range, completeness and consistency validation checks.

The imaging database will be developed separately as part of the web-based systematic image review system (<http://sirs2.neuroimage.co.uk/sirs2/>), only accessible to members of the central team. Data will be stored on secure servers that require user authentication,

archived after the end of the trial, and made available in de-identified format to satisfy data sharing requirements at the end of the trial.

14.8.1 Personal Data

Demographic, clinical, and radiographic personal data will be collected as part of the research, as described above in sections 14.2, 14.3, 14.4, and 14.5. Personal data will be stored by the research team on the Edinburgh Clinical Trials Unit's web server, which can be accessed only by research staff involved in SoSTART. Personal identifiers (e.g. CHI) are encrypted, and the decryption key is stored on the same server. Personal data will be stored for a minimum of five years from the protocol-defined end of trial point (when the minimum retention period has elapsed, trial documentation will not be destroyed without permission from the sponsor).

14.8.2 Transfer of Data

Participants are informed about the need for the trial to access data from the following sources, which involves transfer of their personal to ensure that they are identified correctly: sources such as the Office for National Statistics (ONS) / National Records of Scotland (NRS), the NHS central register, and other registries including those managed by NHS Digital or the Information Services Division in Scotland; the participant's general practitioner; hospitals involved in SoSTART; the participant's relatives, carers, or close personal contacts. Participants consent to the use of information about them and their brain scans to support other research in the future, and that the information may be shared anonymously with other researchers, whatever happens to the participant.

14.8.3 Data Processor

The data processor is the Edinburgh Clinical Trials Unit.

14.8.4 Data Controller

The data controller is The University of Edinburgh/NHS Lothian.

15 STATISTICS AND DATA ANALYSIS

15.1 SAMPLE SIZE CALCULATION

In the only published observational study of full dose OAC after intracranial haemorrhage that reported a composite outcome of all serious vascular events ("ischaemic stroke, systemic embolism, or major bleeding"), the primary outcome at one year had occurred in 27.4% of patients on no antithrombotic drugs, 23.2% on antiplatelet drugs, and 17.1% on full dose OAC.⁴⁴ These differences in event rates would be clinically relevant to patients and are likely to exceed the minimal clinically important difference.¹³⁷ A randomised trial with 90% power to detect a difference in the proportion of participants with the primary outcome of all serious vascular events over one year (27.4% comparator vs 17.1% intervention, $p < 0.05$, two-sided, continuity-corrected chi-squared test) needs 360 participants per group. We plan to recruit at least 60 participants in an internal pilot phase

and at least 190 participants in a safety phase (12% equivalence margin in the outcome of symptomatic intracranial haemorrhage, 1-sided $p=0.025$ and power 90%).

There is insufficient information on modification of treatment effect by radiographic biomarkers of small vessel disease (principally, the presence of brain microbleeds¹¹¹) to calculate the sample size required by the MRI sub-study. However, based on experience in RESTART (ISRCTN71907627), ~60% of participants are expected to take part in the MRI sub-study.

15.2 PROPOSED ANALYSES

15.2.1 Methods

We will write a full Statistical Analysis Plan before follow-up and database lock are complete. The primary analysis will include all randomised participants, irrespective of whether they adhere to the allocated treatment, in the group to which they were allocated.

15.2.2 Missing data

Strenuous efforts will be made to obtain virtually complete outcome data. There will be no imputation for the data with regard to missing values or withdrawals for the statistical summaries and statistical analysis unless otherwise specified. Any required/applicable imputation of data will be specified in the Statistical Analysis Plan.

15.2.3 Brain MRI sub-study

Lobar ICH location,¹⁰³ brain microbleeds,^{111,138} and the combination of these two imaging characteristics suggesting cerebral amyloid angiopathy (CAA) as the underlying cause of ICH^{117,139} may increase the risk of recurrent ICH amongst ICH survivors taking full dose OAC. However, the overall balance between this potential increase in risk of recurrent ICH and the reduction in risk of ischaemic events attributable to full dose OAC has not been tested in a randomised controlled trial of ICH survivors with AF. Therefore, we intend to perform a brain MRI sub-study in at least 60% of the participants in SoSTART (as has been possible in RESTART), to test whether these imaging biomarkers modify the effects of full dose OAC.

16 PHARMACOVIGILANCE

The safety of all participants will be monitored by their usual physicians in secondary or primary care, according to current standard medical practice for patients taking antithrombotic drugs specified in the SPCs (e.g. therapeutic drug monitoring of warfarin using the INR¹⁴⁰ and monitoring renal function in case DOAC dose adjustment is required).

The PI is responsible for the detection and documentation of events meeting the criteria and definitions detailed below after randomisation, which may also be carried out by another suitably qualified physician in the research team at that site, who has up-to-date International Conference on Harmonisation Tripartite Guideline for Good Clinical Practice (GCP) training and who has been delegated this role (co-investigator). Full details of contraindications and side effects that have been reported following administration of the IMP can be found in the relevant SPC (see SPCs listed in a separate document).

Participants will be instructed to contact their Investigator if any symptoms develop at any time after being randomised. All outcomes and relevant AEs will be sought from the time a participant is randomised until death or completion of active trial follow-up (not necessarily including record linkage), and must be recorded in the CRF or AE log. In the case of an AE, the Investigator should initiate the appropriate treatment according to their medical judgment. If an AE occurs that – in the opinion of the investigator – jeopardises the safety of the participant (e.g. major bleeding, contraindication to OAC, or pregnancy), then the IMP should be discontinued.

16.1 DEFINITIONS

An **adverse event** (AE) is any untoward medical occurrence in a clinical trial participant which does not necessarily have a causal relationship with an IMP.

An **adverse reaction** (AR) is any untoward and unintended response to an IMP which is related to any dose administered to that participant.

A **serious adverse event** (SAE) or **serious adverse reaction** (SAR) is any AE or AR that at any dose:

- results in death of the clinical trial participant
- is life threatening[‡]
- requires in-patient hospitalisation[§] or prolongation of existing hospitalisation
- results in persistent or significant disability or incapacity
- consists of a congenital anomaly or birth defect
- results in any other significant medical event not meeting the criteria above

A **suspected unexpected serious adverse reaction** (SUSAR) is any AR that is classified as serious and is suspected to be caused by the IMP, that it is not consistent with the information about the IMP in the SPC or Investigators Brochure.

16.2 ASSESSMENT OF AES AND SAES

Seriousness, causality, severity and expectedness will be assessed by the PI or co-investigator. Investigators are responsible for assessing each AE. The Chief Investigator may not downgrade an event that has been assessed by an Investigator as an SAE or SUSAR, but can upgrade an AE to an SAE, SAR or SUSAR if appropriate.

16.3 IDENTIFYING AES AND SAES

Participants will be asked about the occurrence of AEs/SAEs whenever contact is made with them after randomisation during the trial. Open-ended and non-leading verbal questioning of the participant will be used to enquire about AE/SAE occurrence. Participants will also be asked if they have been admitted to hospital. If there is any doubt

[‡] Life-threatening in the definition of an SAE or SAR refers to an event where the participant was at risk of death at the time of the event. It does not refer to an event which hypothetically might have caused death if it were more severe

[§] Any hospitalisation that was planned before randomisation will not meet SAE criteria. Any hospitalisation that is planned after randomisation will meet the SAE criteria

as to whether a clinical observation is an AE, the event will be recorded. AEs and SAEs may also be identified via information from support departments e.g. laboratories.

16.4 RECORDING AES AND SAES

When an AE/SAE occurs, it is the responsibility of the Investigator, or another suitably qualified physician in the research team who is delegated to record and report AEs/SAEs, to review all documentation (e.g. hospital notes, laboratory and diagnostic reports) related to the event. It is the investigator's responsibility to assess whether an AE is an outcome in the trial, an expected non-fatal complication of intracranial haemorrhage, a known adverse reaction to antithrombotic drugs, a SAE, or a SUSAR (see flowchart below). The Investigator will then record all relevant information in the CRF/AE log and on the SAE form (if the AE meets the criteria of serious) and report it directly to the sponsor. If the AE is detected by central means of follow-up, the TCC will initiate the collection of this information, but enlist the help of local site research staff to acquire the relevant clinical and imaging information. Information to be collected includes dose, type of event, onset date, Investigator assessment of severity and causality, date of resolution as well as treatment required, investigations needed and outcome.

16.4.1 Pre-existing medical conditions

Pre-existing medical conditions (i.e. existed prior to informed consent) should be recorded as medical history and only recorded as adverse events if medically judged to have worsened during the study.

16.4.2 Worsening of the underlying condition during the trial

Medical occurrences or symptoms of deterioration that are expected due to the participant's underlying condition should be recorded in the participant's medical notes and only be recorded as AEs on the AE log if medically judged to have unexpectedly worsened during the study. Events that are consistent with the expected progression of the underlying disease should not be recorded as AEs.

Patients with intracranial haemorrhage experience a variety of events as a result of the natural history of the condition or expected complications of it, including:

- A specific complication such as haematoma growth and/or extension into the ventricles, or perihæmatomal oedema, with consequent symptomatic deterioration
- Hydrocephalus
- Epileptic seizure(s)
- Fever
- Hyperglycaemia
- Raised blood pressure
- Chest infections
- Urinary tract infections
- Other infections including those of soft tissues
- Renal dysfunction
- Painful shoulder syndromes
- Pressure sores
- Spasticity or contractures
- Depression, emotionalism
- Any other expected complications of stroke

16.5 ASSESSMENT OF AES AND SAES

Safety reporting will begin from the time when a participant is randomised. Each AE must be assessed for seriousness, causality, severity and ARs must be assessed for expectedness by the Principal Investigator or another suitably qualified physician in the research team who has been delegated this role. The Chief Investigator (CI) may not downgrade an event that has been assessed by an Investigator as an SAE or SUSAR, but can upgrade an AE to an SAE, SAR or SUSAR if appropriate.

16.5.1 Assessment of seriousness

The Investigator will make an assessment of seriousness as defined above.

16.5.2 Assessment of causality

The Investigator will make an assessment of whether the AE/SAE is likely to be related to the IMP according to the definitions below.

- Unrelated: where an event is not considered to be related to the IMP
- Possibly related: The nature of the event, the underlying medical condition, concomitant medication or temporal relationship make it possible that the AE has a causal relationship to the trial drug.

Where non Investigational Medicinal Products (NIMPs) e.g. rescue/escape drugs are given: if the AE is considered to be related to an interaction between the IMP and the NIMP, or where the AE might be linked to either the IMP or the NIMP but cannot be clearly attributed to either one of these, the event will be considered as an AR. Alternative causes such as natural history of the underlying disease, other risk factors and the temporal relationship of the event to the treatment should be considered and investigated.

16.5.3 Assessment of expectedness

If an event is judged to be an AR, the evaluation of expectedness will be made based on knowledge of the reaction and the relevant product information documented in the SPC. The event may be classed as either:

Expected: the AR is consistent with the toxicity of the IMP listed in the SPC.

Unexpected: the AR is not consistent with the toxicity in the SPC.

16.5.4 Assessment of severity

The Investigator will make an assessment of severity for each AE/SAE/SAR/SUSAR and record this on the CRF or SAE form according to one of the following categories:

Mild: an event that is easily tolerated by the participant, causing minimal discomfort and not interfering with every day activities.

Moderate: an event that is sufficiently discomforting to interfere with normal everyday activities.

Severe: an event that prevents normal everyday activities.

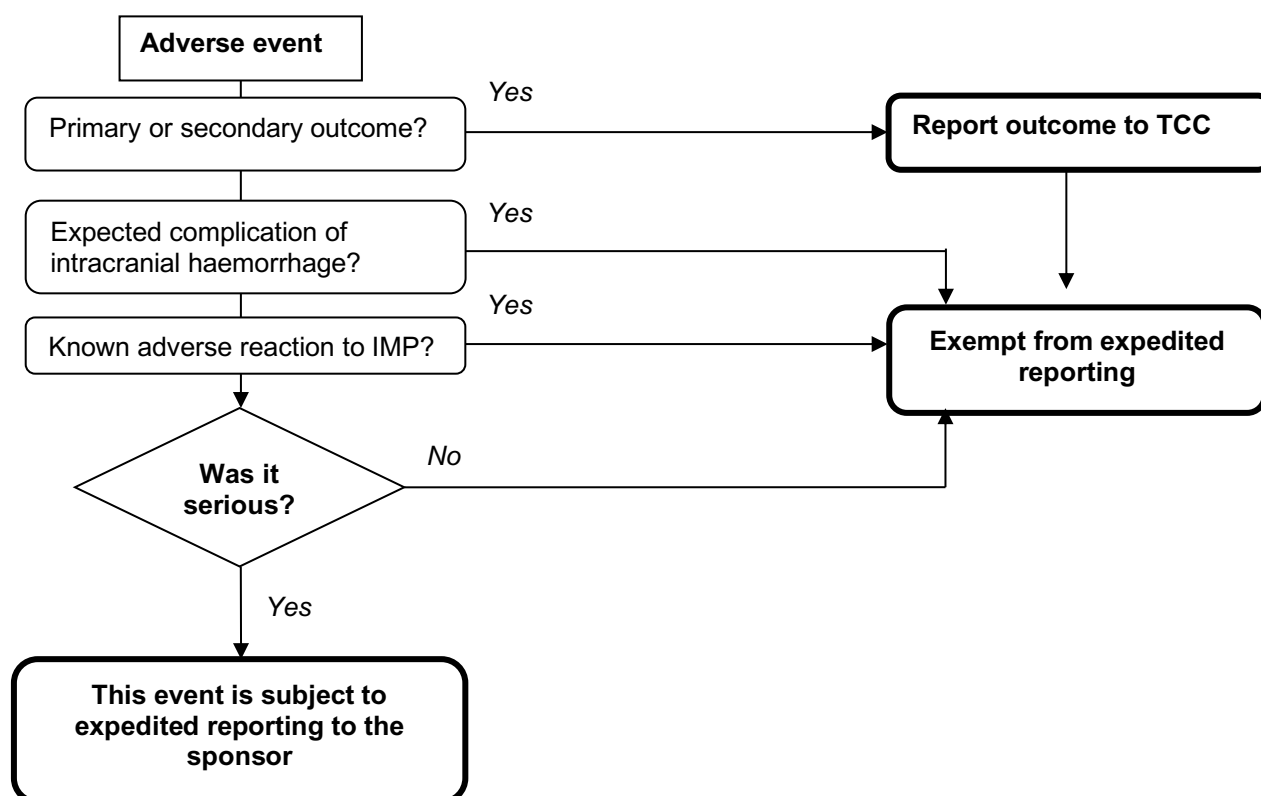
Note: the term 'severe', used to describe the intensity, should not be confused with 'serious' which is a regulatory definition based on participant/event outcome or action criteria. For example, a headache may be severe but not serious, while a minor stroke is serious but may not be severe.

16.5.5 Expected adverse reactions to antithrombotic drugs

ARs that are known to be attributable to antithrombotic drugs are described in the SPCs.

16.6 REPORTING OF SAES/SARS/SUSARS/OUTCOMES IN SOSTART

The expected ARs to the antithrombotic drugs that may be used in SoSTART are well known (see SPCs listed in a separate document), as are the expected non-fatal complications of ICH (see section 16.4.2 above on page 45), so safety assessments in SoSTART are focussed on detecting: primary and secondary outcomes (all of which relate to the safety of antithrombotic drugs in this patient group) and any SAEs and SUSARs that may occur after randomisation. These data will be presented to the DMC and reported in the Development Safety Update Report. Central follow-up will also collect information on hospital admissions and new medications using overlapping sources of ascertainment, which will provide an additional alerting system.



16.7 INVESTIGATORS SHOULD REPORT OUTCOMES TO THE TRIAL CO-ORDINATING CENTRE

Primary and secondary outcomes must be reported to the TCC, at the time of hospital/clinic discharge, or during long-term follow-up if a member of the local research team becomes aware of an event during follow-up.

16.8 INVESTIGATORS MUST REPORT SAES AND SUSARS TO THE TRIAL SPONSOR

The following events occurring after randomisation must be reported to the ACCORD office within 24 hours of the CI or PI becoming aware of the event:

- SUSARs
- SAEs which are neither outcomes, nor known ARs to antithrombotic drugs, nor expected non-fatal complications of intracranial haemorrhage

Once the Investigator becomes aware that an SAE has occurred in a trial participant, the information will be reported to the ACCORD Research Governance & QA Office **within 24 hours**. If the Investigator does not have all information regarding an SAE, they should not wait for this additional information before notifying ACCORD. The SAE report form can be updated when the additional information is received.

The SAE report will provide an assessment of causality and expectedness at the time of the initial report to ACCORD according to sections 16.5.2 and 16.5.3 above.

The SAE form will be transmitted via email to safety@accord.scot. Only forms in a PDF format will be accepted by ACCORD via email. Forms may also be sent by fax to ACCORD on +44 (0)131 242 9447 or may be submitted by hand to the office. Where missing information has not been sent to ACCORD after an initial report, ACCORD will contact the Investigator and request the missing information. The Investigator must respond to these requests in a timely manner.

All reports sent to ACCORD and any follow up information will be retained by the PI in the ISF.

All adverse events for each participant will be recorded on the AE log and will be assigned the appropriate MedDRA Systems Organ Class (SOC) code. AEs will need to be coded by the study team as per EudraCT database requirements.

16.9 INVESTIGATORS SHOULD NOT REPORT OTHER EVENTS TO THE TRIAL CO-ORDINATING CENTRE OR SPONSOR

PIs need not report to the TCC or sponsor any non-fatal AEs that are neither primary/secondary trial outcomes nor SAEs nor SUSARs, and which are expected complications of intracranial haemorrhage.

16.10 REGULATORY REPORTING REQUIREMENTS

ACCORD is responsible for pharmacovigilance reporting on behalf of the co-sponsors (The University of Edinburgh and NHS Lothian).

ACCORD has a legal responsibility to notify the regulatory competent authority and relevant ethics committee (Research Ethics Committee (REC) that approved the trial). Fatal or life threatening SUSARs will be reported no later than 7 calendar days and all other SUSARs will be reported no later than 15 calendar days after ACCORD is first aware of the reaction.

ACCORD (or delegate) will inform Investigators at participating sites of all SUSARs and any other arising safety information.

ACCORD will be responsible for providing safety line listings and assistance; however, it is the responsibility of the Investigator to prepare the Development Safety Update Report. This annual report lists all SARs and SUSARs reported during that time period. The responsibility of submitting the Development Safety Update Report to the regulatory authority and RECs, lies with ACCORD.

An Annual Safety Report/Development Safety Update Report will be submitted, by ACCORD, to the regulatory authorities and RECs listing all SARs and SUSARs.

16.11 FOLLOW UP PROCEDURES

After initially recording an AE or recording and reporting an SAE, the Investigator should make every effort to follow each event until a final outcome can be recorded or reported as necessary. Follow up information on an SAE will be reported to the ACCORD office. If, after follow up, resolution of an event cannot be established, an explanation should be recorded on the CRF or AE log or additional information section of SAE form.

17 PREGNANCY

Women of childbearing potential (i.e. who have not been surgically sterilized or have not been free from menses for >1 year)

- Must have a negative pregnancy test before starting the allocated treatment, and
 - Must not be breast feeding, and
 - Will be asked to use either:
 - True sexual abstinence when this is in line with the preferred and usual lifestyle of the patient. Periodic abstinence (e.g. calendar, ovulation, symptothermal, post-ovulation methods) and withdrawal are not acceptable methods of contraception.
- or*
- Male sterilization (at least 6 months before screening). For female participants, the vasectomised male partner should be their sole partner.
- or one of the following methods:*
- Use of oral (oestrogen and progesterone), injected or implanted hormonal methods of contraception or placement of an intrauterine device (IUD) or intrauterine system (IUS) or other forms of hormonal contraception that have comparable efficacy (failure rate <1%), for example hormone vaginal ring or transdermal hormone contraception.

All methods of contraception must be used in combination with the use of a condom by the male sexual partner for intercourse, from the time of screening until 3 months after discontinuing study treatment.

Male participants must avoid unprotected sex with all sexual partners (by use of condoms) during the study, and for a washout period of 3 months after the last dose of study drug.

Further details can be found here: www.hma.eu/fileadmin/dateien/Human_Medicines/01-About_HMA/Working_Groups/CTFG/2014_09_HMA_CTFG_Contraception.pdf.

Pregnancy is not considered an AE or SAE; however, the Investigator will collect pregnancy information for any female participants or female partners of male participants who become pregnant while participating in the trial. The Investigator will record the information on a Pregnancy Notification Form and submit this to the ACCORD office within 14 days of being made aware of the pregnancy. All pregnant female participants and partners of male participants will be followed up until following the outcome of the pregnancy.

18 TRIAL MANAGEMENT AND OVERSIGHT ARRANGEMENTS

18.1 TRIAL COORDINATING CENTRE

The TCC is responsible for all aspects of the management of SoSTART and is based at the Centre for Clinical Brain Sciences at the University of Edinburgh. Responsibilities include: regulatory submissions and compliance; financial management; monitoring of sites; training; patient information and communication; outcome assessment; data collection systems and data management; statistical analysis; reports and publications and archiving of the trial master file (TMF) in accordance with funder and sponsor requirements. Documentation of these procedures is kept at the TCC.

18.2 TRIAL MANAGEMENT GROUP

The trial will be coordinated by a Trial Management Group, consisting of the grant holders (Chief Investigator and Principal Investigator in Edinburgh), the Assistant Trial Manager, the Trial Manager and other trial staff. This group will be based in the University of Edinburgh Neurosciences Trials Unit at the Centre for Clinical Brain Sciences, affiliated to our UKCRC-registered Edinburgh Clinical Trials Unit (ECTU; www.clinicaltrials.ed.ac.uk). Colleagues in ECTU will provide the trial database, imaging database, and statistical support. This group will meet at least monthly.

The Assistant Trial Manager and the Trial Manager will oversee the trial and will be accountable to the Chief Investigator. The Assistant Trial Manager will be responsible for checking the CRFs for completeness, plausibility and consistency. Any queries will be resolved by the investigator or delegated member of the trial team.

A Delegation Log will be prepared for each site, detailing the responsibilities of each member of staff working on the trial.

18.3 TRIAL STEERING COMMITTEE

A Trial Steering Committee (TSC) has been established to oversee the conduct and progress of the trial (see section 25 below on page 65), check on feasibility, consider whether to proceed from pilot to safety phases and a definitive trial, and reconsider study design. The terms of reference of the Trial Steering Committee, the draft template for reporting and the names and contact details are detailed in CR015 DMC & TSC Charters. The TSC will review the target sample size and adjust it if necessary based on accruing data on: the number of primary outcome events, completeness of follow up, and the enrolment into pre-specified subgroups. The TSC will review outcomes with treatment allocation group masked. The trial may be stopped by the TSC following a recommendation by the DMC, based on their safety and efficacy considerations guided by

the DMC charter. Information about the overall rates of all serious vascular events (for both trial arms combined) will be used together with reasonable assumptions about the effects of full dose OAC on such events in order to assess the plausibility of a net benefit emerging in a definitive trial.

18.4 DATA MONITORING COMMITTEE

A DMC (see section 26 below on page 66), which is independent of the sponsor, oversees the safety of participants in the trial, according to the terms of reference in the DMC Charter, which is available from the TCC. The terms of reference of the Data Monitoring Committee and the names and contact details are detailed in CR0015 DMC & TSC Charters. The DMC Charter will be signed by the appropriate individuals before the trial begins.

During the period of recruitment into the trial, interim analyses of the baseline and follow up data will be supplied, in strict confidence, to the chairman of the DMC, along with any other analyses that the committee may request. In the light of these analyses, the DMC will advise the chairman of the TSC if, in their view, the randomised comparisons have provided both (i) 'proof beyond reasonable doubt' that for all, or some, the treatment is clearly indicated or clearly contra-indicated and (ii) evidence that might reasonably be expected to materially influence future patient management. Appropriate criteria of proof beyond reasonable doubt cannot be specified precisely, but the DMC will work on the principle that the Haybittle-Peto boundary of at least three standard errors' difference (i.e. $p < 0.001$) in an interim analysis of a major outcome event (e.g. the primary outcome) may be needed to justify halting, or modifying, a trial before the planned completed recruitment. This criterion has the practical advantage that the exact number of interim analyses would be of little importance, and so no fixed schedule is proposed. Following such a report from the DMC, the TSC will decide whether to modify entry to the trial (or seek extra data). Unless this happens however, the TSC, the collaborators and central administrative staff will remain ignorant of the interim results.

18.5 INSPECTION OF RECORDS

Investigators and institutions involved in the trial will permit trial related monitoring and audits on behalf of the sponsor, REC review, and regulatory inspection(s). In the event of an audit or monitoring, the PI agrees to allow the representatives of the sponsor direct access to all trial records and source documentation. In the event of regulatory inspection, the PI agrees to allow inspectors direct access to all trial records and source documentation.

18.6 RISK ASSESSMENT

A study specific risk assessment will be performed by representatives of the co-sponsors, ACCORD monitors and the QA group, in accordance with ACCORD governance and sponsorship SOPs. Input will be sought from the Chief Investigator or designee. The outcomes of the risk assessment will form the basis of the monitoring plans and audit plans. The risk assessment outcomes will also indicate which risk adaptations could be incorporated into trial design.

18.7 TRIAL MONITORING AND AUDIT

ACCORD clinical trial monitors, or designees, will perform monitoring activities in accordance with the study monitoring plan. This will involve on-site visits and remote monitoring activities as necessary. ACCORD QA personnel, or designees, will perform study audits in accordance with the study audit plan. This will involve investigator site audits, study management audits and facility (including third parties) audits as necessary.

19 GOOD CLINICAL PRACTICE

19.1 ETHICAL CONDUCT

The trial will be conducted in accordance with the principles of the International Conference on Harmonisation Tripartite Guideline for Good Clinical Practice (ICH GCP). Before the trial can begin, all required approvals will be obtained and any conditions of approvals will be met.

19.2 REGULATORY COMPLIANCE

The trial will not begin until a Clinical Trial Authorisation (CTA) is obtained from the appropriate Regulatory Authority. The protocol and study conduct will comply with the Medicines for Human Use (Clinical Trials) Regulations 2004, as amended.

19.3 PRINCIPAL INVESTIGATOR RESPONSIBILITIES

The PI is responsible for the overall conduct of the trial at the site and compliance with the protocol and any protocol amendments. In accordance with the principles of GCP, the following areas listed in this section are also the responsibility of the Investigator. Responsibilities may be delegated to an appropriate member of trial site staff and recorded on the delegation log.

Delegated tasks must be documented on a Delegation Log and signed by all those named on the list prior to undertaking applicable study-related procedures.

19.3.1 Confirming participant eligibility and obtaining informed consent

The PI or co-investigator must confirm in writing in the paper and/or electronic medical records that a patient fulfils the eligibility criteria. The decision of a participant to participate in clinical research is voluntary and should be based on a clear understanding of what is involved. Participants must receive adequate oral and written information – appropriate PILs and ICFs will be provided. The oral explanation to the participant will be performed by the PI, co-investigator or qualified delegated person, and must cover all the elements specified in the PILs and ICF. The participant will be informed and agree to their medical records being inspected by regulatory authorities and representatives of the sponsor(s) but understand that their name will not be disclosed outside the hospital or TCC. The participant must be given every opportunity to clarify any points they do not understand and, if necessary, ask for more information. The participant must be given sufficient time to consider the information provided. It should be emphasised that the participant may withdraw their consent to participate at any time without their medical care being affected. The Investigator or delegated member of the trial team and the participant will sign and

date the ICF to confirm that consent has been obtained. The original consent form should be filed in the ISF, the participant will receive a copy of this document and another copy should be filed in the participant's medical notes along with a copy of the PILs used. The PI is responsible for ensuring informed consent is obtained before any protocol specific procedures are carried out (including brain MRI if in the sub-study). The PI or co-investigator with delegated responsibility must prescribe antithrombotic drugs (if the participant is randomised to a policy of starting them). Another member of the research team may be delegated the responsibility for identifying suitable patients, obtaining consent and randomising participants.

19.3.2 Trial site staff

Investigators must be familiar with the IMP, protocol and the trial requirements. It is the PI's responsibility to ensure that all staff assisting with the trial are adequately informed about the IMP, protocol and their trial related duties.

19.3.3 Data recording

The PI is responsible for the quality of the data recorded in the CRF at each site. The source data plan identifies which source data correspond to CRF data and states which data are recorded directly into the CRF.

19.3.4 Investigator documentation

Before beginning the trial, each PI will be asked to provide particular essential documents to the ACCORD Research Governance & QA Office, including but not limited to:

- An original signed PI's Declaration (as part of the Clinical Trial Agreement documents);
- Curriculum vitae (CV) signed and dated by the investigator in the previous two years indicating that it is accurate and current.
- ACCORD will ensure all other documents required by ICH GCP are retained in a Trial Master File (TMF) or Sponsor File, where required. The Principal Investigator will ensure that the required documentation is available in local Investigator Site files ISFs. Under certain circumstances, the TMF responsibilities may be delegated to the research team by ACCORD.

19.3.5 GCP training

All trial staff must hold evidence of appropriate GCP training in keeping with each site's local R&D policy.

19.3.6 Confidentiality

All laboratory specimens, evaluation forms, reports, and other records must be identified in a manner designed to maintain participant confidentiality. All records must be kept in a secure storage area with limited access. Clinical identifiable information will not be released without the written permission of the participant. The investigators and trial site staff involved with this trial may not disclose or use for any purpose other than

performance of the trial, any data, record, or other unpublished, confidential identifiable information disclosed to those individuals for the purpose of the trial. Prior written agreement from the sponsor or its designee must be obtained for the disclosure of any said confidential identifiable information to other parties.

20 ACCESS TO DATA

All Investigators and trial site staff involved with this trial must comply with the requirements of local data protection legislation (including the General Data Protection Regulation and Data Protection Act) with regard to the collection, storage, processing and disclosure of personal information and will uphold the Act's core principles. Access to collated identifiable participant data will be restricted to those clinicians treating the participants, representatives of the co-sponsors and representatives of regulatory authorities. Computers used to collate the data must have limited access measures via user names and passwords. Published results will not contain any personal data that could allow identification of individual participants. Members of the TCC will have access to the final trial data set. Ownership of the data arising from this trial resides with the TCC. Decisions to release the anonymised final trial data set to third parties will be made by the Chief Investigator, with advice from the TSC.

21 TRIAL CONDUCT RESPONSIBILITIES

21.1 PROTOCOL AMENDMENTS

Any changes in research activity, except those necessary to remove an apparent, immediate hazard to the participant in the case of an urgent safety measure, must be reviewed and approved by the Chief Investigator.

Proposed amendments will be submitted to the Sponsor for classification and authorisation.

Amendments to the protocol must be submitted in writing to the appropriate REC, regulatory authorities and local R&D for approval before participants are enrolled according to an amended protocol. The TCC will be responsible for disseminating information about protocol amendments to investigators, ethics committees, other regulators, trial participants, and trial registries.

21.2 PROTOCOL NON-COMPLIANCE

21.2.1 Definitions

Deviation - Any change, divergence, or departure from the study design, procedures defined in the protocol or GCP that does not significantly affect a subjects rights, safety, or well-being, or study outcomes.

Violation - A deviation that may potentially significantly impact the completeness, accuracy, and/or reliability of the study data or that may significantly affect a subject's rights, safety, or well-being

21.2.2 Protocol Waivers

Prospective protocol deviations, i.e. protocol waivers, will not be approved by the sponsors and therefore will not be implemented, except where necessary to eliminate an immediate hazard to study participants. If this necessitates a subsequent protocol amendment, this should be submitted to the REC, Regulatory Authority and local R&D for review and approval if appropriate.

21.2.3 Management of Deviations and Violations

Protocol deviations will be recorded by each site in a protocol deviation log and logs will be submitted to the sponsors every 3 months at the latest. Each protocol violation will be reported to the sponsor within 3 days of becoming aware of the violation. Deviation logs / violation forms will be transmitted via email to QA@accord.scot. Only forms in a pdf format will be accepted by ACCORD via email. Forms may also be sent by fax to ACCORD on +44 (0)131 242 9447 or may be submitted by hand to the office. Where missing information has not been sent to ACCORD after an initial report, ACCORD will contact the Investigator and request the missing information. The Investigator must respond to these requests in a timely manner

21.3 URGENT SAFETY MEASURES

The Investigator may implement a deviation from or change to the protocol to eliminate an immediate hazard to trial participants without prior approval from the REC and the MHRA. This is defined as an urgent safety measure and the investigator must contact the Clinical Trial Unit at the MHRA and discuss the issue with a medical assessor immediately (+44 (0) 20 3080 6456).

The Investigator will then notify the MHRA (clintrialhelpline@mhra.gsi.gov.uk), the REC and ACCORD, in writing of the measures taken and the reason for the measures within 3 days by submitting a substantial amendment.

21.4 SERIOUS BREACH REQUIREMENTS

A serious breach is a breach which is likely to effect to a significant degree: (a) the safety or physical or mental integrity of the participants of the trial, or (b) the scientific value of the trial.

If a potential serious breach is identified by the Chief investigator, PI or delegates, the co-sponsors (QA@accord.scot) must be notified within 24 hours. It is the responsibility of the co-sponsors to assess the impact of the breach on the scientific value of the trial, to determine whether the incident constitutes a serious breach and report to regulatory authorities and REC as necessary.

21.5 TRIAL RECORD RETENTION

All trial documentation will be kept for a minimum of five years from the protocol defined end of trial point. When the minimum retention period has elapsed, trial documentation will not be destroyed without permission from the sponsor.

21.6 END OF TRIAL

The end of the trial is defined as the last active follow-up received (not including record linkage).

The Investigators and/or the trial steering committee and/or the co-sponsor(s) have the right at any time to terminate the study for clinical or administrative reasons.

The end of the study will be reported to the REC, Regulatory Authority, R&D Office(s) and co-sponsors within 90 days, or 15 days if the study is terminated prematurely. The Investigators will inform participants of the premature study closure and ensure that the appropriate follow up is arranged for all participants involved. End of study notification will be reported to the co-sponsors via email to resgov@accord.scot.

In accordance with ACCORD SOP CR011, a Clinical Study Report (CSR) will be provided to the Sponsor (QA@accord.scot), and REC within 1 year of the end of the study. Upon completion of the study, the Investigator will upload clinical trial results onto the EudraCT database on behalf of the Sponsor.

The Investigator will submit a short confirmatory e-mail to the MHRA (CT.Submission@mhra.gsi.gov.uk) once the result-related information has been uploaded to EudraCT, with 'End of trial: result-related information: EudraCT 2016-004121-16' as the subject line. The Sponsor(s) will be copied in this e-mail (QA@accord.scot). It should be noted that you will not get an acknowledgment e-mail or letter from the MHRA.

21.7 CONTINUATION OF DRUG FOLLOWING THE END OF TRIAL

Antithrombotic drugs may be continued indefinitely after the end of the trial follow-up period. There are no provisions for ancillary care or care after the trial for participants.

21.8 INSURANCE AND INDEMNITY

The co-sponsors are responsible for ensuring proper provision has been made for insurance or indemnity to cover their liability and the liability of the Chief Investigator and staff.

The following arrangements are in place to fulfil the co-sponsors' responsibilities:

- The protocol has been designed by the Chief Investigator and researchers employed by the University. The University has insurance in place (which includes no-fault compensation) for negligent harm caused by poor protocol design by the Chief Investigator and researchers employed by the University.
- Sites participating in the trial will be liable for clinical negligence and other negligent harm to individuals taking part in the trial and covered by the duty of care owed to them by the sites concerned. The co-sponsors require individual sites participating in the trial to arrange for their own insurance or indemnity in respect of these liabilities.

- Sites that are part of the United Kingdom's National Health Service will have the benefit of NHS Indemnity.
- Sites outside the United Kingdom will be responsible for arranging their own indemnity or insurance for their participation in the trial, as well as for compliance with local law applicable to their participation in the trial.
- The manufacturer supplying IMP has accepted limited liability related to the manufacturing and original packaging of the trial drug and to the losses, damages, claims or liabilities incurred by trial participants based on known or unknown adverse events which arise out of the manufacturing and original packaging of the trial drug, but not where there is any modification to the trial drug (including without limitation re-packaging and masking).

22 DISSEMINATION POLICY

22.1 COMMUNICATION

The investigators and sponsor will communicate trial results to participants, health care professionals, the public, and other relevant groups (e.g., via publication, reporting in results databases, or other data-sharing arrangements), including any publication restrictions, using the most up-to-date contact information available in the trial database.

22.2 AUTHORSHIP POLICY

On completion of the trial, the trial data will be analysed and tabulated, and a clinical trial report will be prepared in accordance with International Conference on Harmonisation guidelines. The success of this trial depends entirely on the collaboration of a large number of doctors, nurses, other health professionals, patients and relatives. Those included in the Delegation Logs will be included in any listing of collaborators. For this reason the credit for the main results will be given, not exclusively to the central trial coordinators, but to all wholehearted collaborators in the trial. The primary trial publication will be drafted by a writing committee whose membership has been approved by the TSC. The manuscript must be approved by the TSC before submission for publication.

22.3 PUBLICATION

The Clinical Study Report (CSR) will be submitted to the Sponsor and REC within 1 year of the end of the study. Where acceptable, a published journal article may be submitted as the CSR. The Chief Investigator will provide the CSR to ACCORD, for review, prior to finalization. The clinical study report may be used for publication and presentation at scientific meetings. Investigators have the right to publish orally or in writing the results of the study. The results of the study, together with other mandated information, will be uploaded to the European clinical trials database within 1 year of the end of the study.

Summaries of results will also be made available to Investigators for dissemination within their clinics (where appropriate and according to their discretion).

The results will also be disseminated to participants and the public via the trial website and social media. Summaries of results will also be made available to PIs for dissemination within their clinics (where appropriate and according to their discretion).

23 PEER REVIEW

An outline of this protocol was peer reviewed by Chest Heart & Stroke Scotland and the British Heart Foundation when submitted as a grant application. This protocol has been reviewed by the co-sponsors and the TSC.

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25 APPENDIX 1: Trial Steering Committee

- Colin Baigent (independent chairperson)
Professor of epidemiology and Director of the MRC Population Health Research Unit, University of Oxford
- Rustam Al-Shahi Salman (chief investigator)
*Professor of clinical neurology, University of Edinburgh
Honorary consultant neurologist, NHS Lothian*
- Kasia Adamczuk (trial manager)
CCBS, University of Edinburgh
- Karen Innes (senior trial manager)
FOCUS and RESTART trial manager, University of Edinburgh
- Colin Oliver (non-medical 'lay' member)
Lay member, Edinburgh
- Steff Lewis (trial statistician)
Personal chair in medical statistics, University of Edinburgh
- David E. Newby
BHF Professor of cardiology, University of Edinburgh
- Gregory Y.H. Lip
Professor of Cardiovascular Medicine, University of Birmingham, and Honorary Consultant Cardiologist, Sandwell & West Birmingham Hospitals NHS Trust
- Martin S. Dennis
Professor of stroke medicine, University of Edinburgh
- Adrian Parry-Jones
*NIHR Clinician Scientist, University of Manchester
Honorary consultant neurologist, Salford*
- Dan Lasserson
Professor of Ambulatory Care, University of Birmingham
- Sponsor representative
ACCORD, University of Edinburgh & NHS Lothian
- Phil M. White
Professor of interventional and diagnostic neuroradiology, University of Newcastle
- Shannon Amoils (funding representative)
Senior Research Adviser, British Heart Foundation



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26 APPENDIX 2: Data Monitoring Committee

- Dr John Bamford (chairperson)
Consultant neurologist and Clinical Associate Professor, Leeds General Infirmary, Leeds
- Prof Jane Armitage
Professor of Clinical Trials and Epidemiology, CTSU, University of Oxford
- Prof Jonathan Emberson
Professor of Medical Statistics and Epidemiology, CTSU, University of Oxford
- Professor Gabriël Rinkel
Professor of neurology, University Medical Center, Utrecht, The Netherlands
- Professor Gordon Lowe
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